

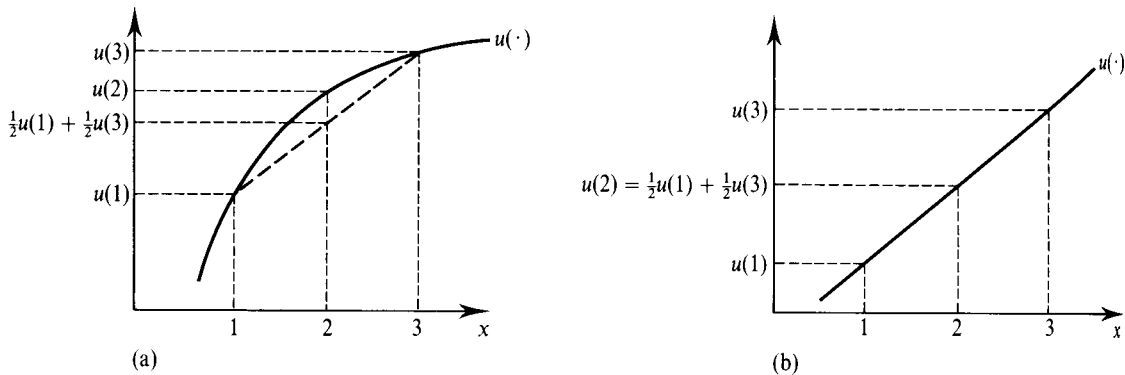


Figure 6.C.2 Risk aversion (a) and risk neutrality (b).

context of expected utility theory, we see that *risk aversion is equivalent to the concavity of $u(\cdot)$* and that strict risk aversion is equivalent to the strict concavity of $u(\cdot)$. This makes sense. Strict concavity means that the marginal utility of money is decreasing. Hence, at any level of wealth x , the utility gain from an extra dollar is smaller than (the absolute value of) the utility loss of having a dollar less. It follows that a risk of gaining or losing a dollar with even probability is not worth taking. This is illustrated in Figure 6.C.2(a); in the figure we consider a gamble involving the gain or loss of 1 dollar from an initial position of 2 dollars. The (v.N-M) utility of this gamble, $\frac{1}{2}u(1) + \frac{1}{2}u(3)$, is strictly less than that of the initial certain position $u(2)$.

For a risk-neutral expected utility maximizer, (6.C.2) must hold with *equality* for all $F(\cdot)$. Hence, the decision maker is risk neutral if and only if the Bernoulli utility function of money $u(\cdot)$ is linear. Figure 6.C.2(b) depicts the (v.N-M) utility associated with the previous gamble for a risk neutral individual. Here the individual is indifferent between the gambles that yield a mean wealth level of 2 dollars and a certain wealth of 2 dollars. Definition 6.C.2 introduces two useful concepts for the analysis of risk aversion.

Definition 6.C.2: Given a Bernoulli utility function $u(\cdot)$ we define the following concepts:

- (i) The *certainty equivalent* of $F(\cdot)$, denoted $c(F, u)$, is the amount of money for which the individual is indifferent between the gamble $F(\cdot)$ and the certain amount $c(F, u)$; that is,

$$u(c(F, u)) = \int u(x) dF(x). \quad (6.C.3)$$

- (ii) For any fixed amount of money x and positive number ε , the *probability premium* denoted by $\pi(x, \varepsilon, u)$, is the excess in winning probability over fair odds that makes the individual indifferent between the certain outcome x and a gamble between the two outcomes $x + \varepsilon$ and $x - \varepsilon$. That is

$$u(x) = \left(\frac{1}{2} + \pi(x, \varepsilon, u)\right)u(x + \varepsilon) + \left(\frac{1}{2} - \pi(x, \varepsilon, u)\right)u(x - \varepsilon). \quad (6.C.4)$$

These two concepts are illustrated in Figure 6.C.3. In Figure 6.C.3(a), we exhibit the geometric construction of $c(F, u)$ for an even probability gamble between 1 and 3 dollars. Note that $c(F, u) < 2$, implying that some expected return is traded for certainty. The satisfaction of the inequality $c(F, u) \leq \int x dF(x)$ for all $F(\cdot)$ is, in fact,

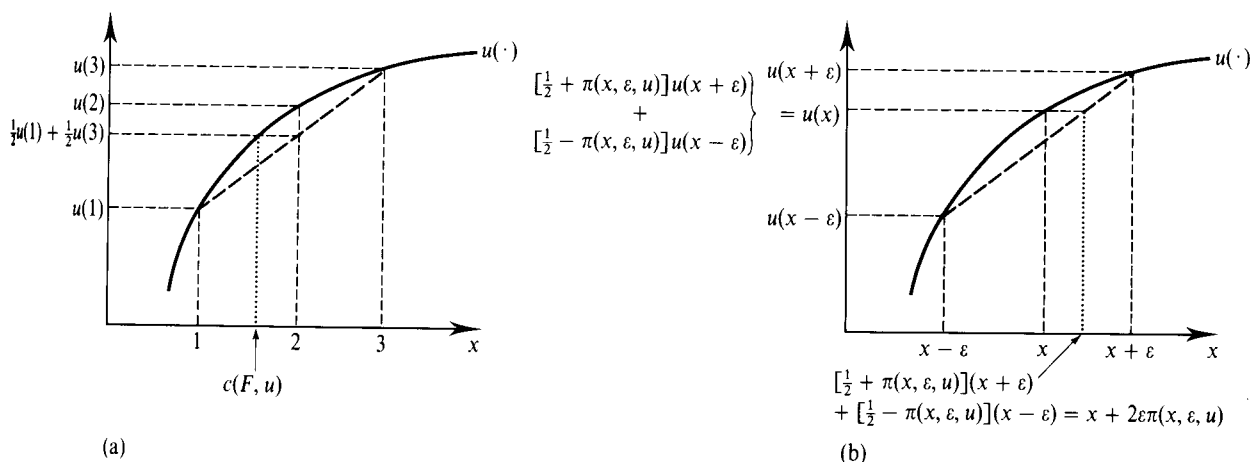


Figure 6.C.3 The certainty equivalent (a) and the probability premium (b).

equivalent to the decision maker being a risk averter. To see this, observe that since $u(\cdot)$ is nondecreasing, we have

$$c(F, u) \leq \int x dF(x) \Leftrightarrow u(c(F, u)) \leq u\left(\int x dF(x)\right) \Leftrightarrow \int u(x) dF(x) \leq u\left(\int x dF(x)\right),$$

where the last $\Leftrightarrow$ follows from the definition of $c(F, u)$.

In Figure 6.C.3(b), we exhibit the geometric construction of $\pi(x, \varepsilon, u)$. We see that $\pi(x, \varepsilon, u) > 0$; that is, better than fair odds must be given for the individual to accept the risk. In fact, the satisfaction of the inequality $\pi(x, \varepsilon, u) \geq 0$ for all x and $\varepsilon > 0$ is also equivalent to risk aversion (see Exercise 6.C.3).

These points are formally summarized in Proposition 6.C.1.

Proposition 6.C.1: Suppose a decision maker is an expected utility maximizer with a Bernoulli utility function $u(\cdot)$ on amounts of money. Then the following properties are equivalent:

- (i) The decision maker is risk averse.
- (ii) $u(\cdot)$ is concave.¹⁶
- (iii) $c(F, u) \leq \int x dF(x)$ for all $F(\cdot)$.
- (iv) $\pi(x, \varepsilon, u) \geq 0$ for all x, ε .

Examples 6.C.1 to 6.C.3 illustrate the use of the risk aversion concept.

Example 6.C.1: Insurance. Consider a strictly risk-averse decision maker who has an initial wealth of w but who runs a risk of a loss of D dollars. The probability of the loss is π . It is possible, however, for the decision maker to buy insurance. One unit of insurance costs q dollars and pays 1 dollar if the loss occurs. Thus, if α units of insurance are bought, the wealth of the individual will be $w - \alpha q$ if there is no loss and $w - \alpha q - D + \alpha$ if the loss occurs. Note, for purposes of later discussion, that the decision maker's expected wealth is then $w - \pi D + \alpha(\pi - q)$. The decision maker's problem is to choose the optimal level of α . His utility maximization problem is

16. Recall that if $u(\cdot)$ is twice differentiable then concavity is equivalent to $u''(x) \leq 0$ for all x .

therefore

$$\text{Max}_{\alpha \geq 0} (1 - \pi)u(w - \alpha q) + \pi u(w - \alpha q - D + \alpha).$$

If α^* is an optimum, it must satisfy the first-order condition:

$$-q(1 - \pi)u'(w - \alpha^*q) + \pi(1 - q)u'(w - D + \alpha^*(1 - q)) \leq 0,$$

with equality if $\alpha^* > 0$.

Suppose now that the price q of one unit of insurance is *actuarially fair* in the sense of it being equal to the expected cost of insurance. That is, $q = \pi$. Then the first-order condition requires that

$$u'(w - D + \alpha^*(1 - \pi)) - u'(w - \alpha^*\pi) \leq 0,$$

with equality if $\alpha^* > 0$.

Since $u'(w - D) > u'(w)$, we must have $\alpha^* > 0$, and therefore

$$u'(w - D + \alpha^*(1 - \pi)) = u'(w - \alpha^*\pi).$$

Because $u'(\cdot)$ is strictly decreasing, this implies

$$w - D + \alpha^*(1 - \pi) = w - \alpha^*\pi,$$

or, equivalently,

$$\alpha^* = D.$$

Thus, if insurance is *actuarially fair*, the decision maker insures completely. The individual's final wealth is then $w - \pi D$, regardless of the occurrence of the loss.

This proof of the complete insurance result uses first-order conditions, which is instructive but not really necessary. Note that if $q = \pi$, then the decision maker's expected wealth is $w - \pi D$ for any α . Since setting $\alpha = D$ allows him to reach $w - \pi D$ with certainty, the definition of risk aversion directly implies that this is the optimal level of α . ■

Example 6.C.2: Demand for a Risky Asset. An asset is a divisible claim to a financial return in the future. Suppose that there are two assets, a safe asset with a return of 1 dollar per dollar invested and a risky asset with a random return of z dollars per dollar invested. The random return z has a distribution function $F(z)$ that we assume satisfies $\int z dF(z) > 1$; that is, its mean return exceeds that of the safe asset.

An individual has initial wealth w to invest, which can be divided in any way between the two assets. Let α and β denote the amounts of wealth invested in the risky and the safe asset, respectively. Thus, for any realization z of the random return, the individual's *portfolio* (α, β) pays $\alpha z + \beta$. Of course, we must also have $\alpha + \beta = w$.

The question is how to choose α and β . The answer will depend on $F(\cdot)$, w , and the Bernoulli utility function $u(\cdot)$. The utility maximization problem of the individual is

$$\begin{aligned} \text{Max}_{\alpha, \beta \geq 0} \quad & \int u(\alpha z + \beta) dF(z) \\ \text{s.t.} \quad & \alpha + \beta = w. \end{aligned}$$

Equivalently, we want to maximize $\int u(w + \alpha(z - 1)) dF(z)$ subject to $0 \leq \alpha \leq w$. If

α^* is optimal, it must satisfy the Kuhn–Tucker first-order conditions:¹⁷

$$\phi(\alpha^*) = \int u'(w + \alpha^*[z - 1])(z - 1) dF(z) \begin{cases} \leq 0 & \text{if } \alpha^* < w, \\ \geq 0 & \text{if } \alpha^* > 0. \end{cases}$$

Note that $\int z dF(z) > 1$ implies $\phi(0) > 0$. Hence, $\alpha^* = 0$ cannot satisfy this first-order condition. We conclude that the optimal portfolio has $\alpha^* > 0$. The general principle illustrated in this example, is that *if a risk is actuarially favorable, then a risk averter will always accept at least a small amount of it.*

This same principle emerges in Example 6.C.1 if insurance is not actuarially fair. In Exercise 6.C.1, you are asked to show that if $q > \pi$, then the decision maker will not fully insure (i.e., will accept some risk). ■

Example 6.C.3: General Asset Problem. In the previous example, we could define the utility $U(\alpha, \beta)$ of the portfolio (α, β) as $U(\alpha, \beta) = \int u(\alpha z + \beta) dF(z)$. Note that $U(\cdot)$ is then an increasing, continuous, and concave utility function. We now discuss an important generalization. We assume that we have N assets (one of which may be the safe asset) with asset n giving a return of z_n per unit of money invested. These returns are jointly distributed according to a distribution function $F(z_1, \dots, z_N)$. The utility of holding a *portfolio* of assets $(\alpha_1, \dots, \alpha_N)$ is then

$$U(\alpha_1, \dots, \alpha_N) = \int u(\alpha_1 z_1 + \dots + \alpha_N z_N) dF(z_1, \dots, z_N).$$

This utility function for portfolios, defined on $\mathbb{R}_+^N$, is also increasing, continuous, and concave (see Exercise 6.C.4). This means that, formally, we can treat assets as the usual type of commodities and apply to them the demand theory developed in Chapters 2 and 3. Observe, in particular, how risk aversion leads to a convex indifference map for portfolios. ■

Suppose that the lotteries pay in vectors of physical goods rather than in money. Formally, the space of outcomes is then the consumption set $\mathbb{R}_+^L$ (all the previous discussion can be viewed as the special case in which there is a single good). In this more general setting, the concept of risk aversion given by Definition 6.C.1 is perfectly well defined. Furthermore, if there is a Bernoulli utility function $u: \mathbb{R}_+^L \rightarrow \mathbb{R}$, then risk aversion is still equivalent to the concavity of $u(\cdot)$. Hence, we have here another justification for the convexity assumption of Chapter 3: Under the assumptions of the expected utility theorem, the convexity of preferences for perfectly certain amounts of the physical commodities must hold if for any lottery with commodity payoffs the individual always prefers the certainty of the mean commodity bundle to the lottery itself.

In Exercise 6.C.5, you are asked to show that if preferences over lotteries with commodity payoffs exhibit risk aversion, then, at given commodity prices, the induced preferences on money lotteries (where consumption decisions are made after the realization of wealth) are also risk averse. Thus, in principle, it is possible to build the theory of risk aversion on the more primitive notion of lotteries over the final consumption of goods.

17. The objective function is concave in α because the concavity of $u(\cdot)$ implies that $\int u''(w + \alpha(z - 1))(z - 1)^2 dF(x) \leq 0$.

The Measurement of Risk Aversion

Now that we know what it means to be risk averse, we can try to measure the extent of risk aversion. We begin by defining one particularly useful measure and discussing some of its properties.

Definition 6.C.3: Given a (twice-differentiable) Bernoulli utility function $u(\cdot)$ for money, the *Arrow–Pratt coefficient of absolute risk aversion* at x is defined as $r_A(x) = -u''(x)/u'(x)$.

The Arrow–Pratt measure can be motivated as follows: We know that risk neutrality is equivalent to the linearity of $u(\cdot)$, that is, to $u''(x) = 0$ for all x . Therefore, it seems logical that the degree of risk aversion be related to the *curvature* of $u(\cdot)$. In Figure 6.C.4, for example, we represent two Bernoulli utility functions $u_1(\cdot)$ and $u_2(\cdot)$ normalized (by choice of origin and units) to have the same utility and marginal utility values at wealth level x . The certainty equivalent for a small risk with mean x is smaller for $u_2(\cdot)$ than for $u_1(\cdot)$, suggesting that risk aversion increases with the curvature of the Bernoulli utility function at x . One possible measure of curvature of the Bernoulli utility function $u(\cdot)$ at x is $u''(x)$. However, this is not an adequate measure because it is not invariant to positive linear transformations of the utility function. To make it invariant, the simplest modification is to use $u''(x)/u'(x)$. If we change sign so as to have a positive number for an increasing and concave $u(\cdot)$, we get the Arrow–Pratt measure.

A more precise motivation for $r_A(x)$ as a measure of the degree of risk aversion can be obtained by considering a fixed wealth x and studying the behavior of the probability premium $\pi(x, \varepsilon, u)$ as $\varepsilon \rightarrow 0$ [for simplicity, we write it as $\pi(\varepsilon)$]. Differentiating the identity (6.C.4) that defines $\pi(\cdot)$ twice with respect to ε (assume that $\pi(\cdot)$ is differentiable), and evaluating at $\varepsilon = 0$, we get $4\pi'(0)u'(x) + u''(x) = 0$. Hence

$$r_A(x) = 4\pi'(0).$$

Thus, $r_A(x)$ measures the rate at which the probability premium increases at certainty with the small risk measured by ε .¹⁸ As we go along, we will find additional related interpretations of the Arrow–Pratt measure.

18. For a similar derivation relating $r_A(\cdot)$ to the rate of change of the certainty equivalent with respect to a small increase in a small risk around certainty, see Exercise 6.C.20.

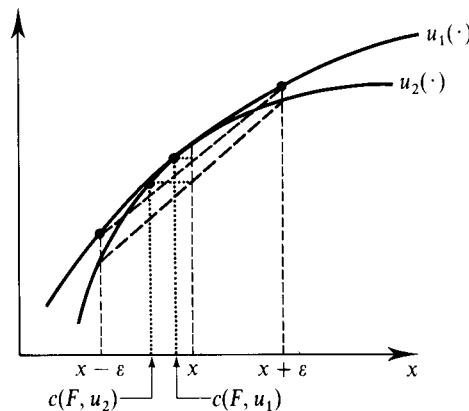


Figure 6.C.4
Differing degrees of risk aversion.

Note that, up to two integration constants, the utility function $u(\cdot)$ can be recovered from $r_A(\cdot)$ by integrating twice. The integration constants are irrelevant because the Bernoulli utility is identified only up to two constants (origin and units). Thus, the Arrow–Pratt risk aversion measure $r_A(\cdot)$ fully characterizes behavior under uncertainty.

Example 6.C.4: Consider the utility function $u(x) = -e^{-ax}$ for $a > 0$. Then $u'(x) = ae^{-ax}$ and $u''(x) = -a^2e^{-ax}$. Therefore, $r_A(x, u) = a$ for all x . It follows from the observation just made that the general form of a Bernoulli utility function with an Arrow–Pratt measure of absolute risk aversion equal to the constant $a > 0$ at all x is $u(x) = -\alpha e^{-ax} + \beta$ for some $\alpha > 0$ and β . ■

Once we are equipped with a measure of risk aversion, we can put it to use in comparative statics exercises. Two common situations are the comparisons of risk attitudes across individuals with different utility functions and the comparison of risk attitudes for one individual at different levels of wealth.

Comparisons across individuals

Given two Bernoulli utility functions $u_1(\cdot)$ and $u_2(\cdot)$, when can we say that $u_2(\cdot)$ is unambiguously *more risk averse than* $u_1(\cdot)$? Several possible approaches to a definition seem plausible:

- (i) $r_A(x, u_2) \geq r_A(x, u_1)$ for every x .
- (ii) There exists an increasing concave function $\psi(\cdot)$ such that $u_2(x) = \psi(u_1(x))$ at all x ; that is, $u_2(\cdot)$ is a concave transformation of $u_1(\cdot)$. [In other words, $u_2(\cdot)$ is “more concave” than $u_1(\cdot)$.]
- (iii) $c(F, u_2) \leq c(F, u_1)$ for any $F(\cdot)$.
- (iv) $\pi(x, \varepsilon, u_2) \geq \pi(x, \varepsilon, u_1)$ for any x and ε .
- (v) Whenever $u_2(\cdot)$ finds a lottery $F(\cdot)$ at least as good as a riskless outcome $\bar{x}$, then $u_1(\cdot)$ also finds $F(\cdot)$ at least as good as $\bar{x}$. That is, $\int u_2(x) dF(x) \geq u_2(\bar{x})$ implies $\int u_1(x) dF(x) \geq u_1(\bar{x})$ for any $F(\cdot)$ and $\bar{x}$.¹⁹

In fact, these five definitions are equivalent.

Proposition 6.C.2: Definitions (i) to (v) of the *more-risk-averse-than* relation are equivalent.

Proof: We will not give a complete proof. (You are asked to establish some of the implications in Exercises 6.C.6 and 6.C.7.) Here we will show the equivalence of (i) and (ii) under differentiability assumptions.

Note, first that we always have $u_2(x) = \psi(u_1(x))$ for some increasing function $\psi(\cdot)$; this is true simply because $u_1(\cdot)$ and $u_2(\cdot)$ are ordinally identical (more money is preferred to less). Differentiating, we get

$$u_2'(x) = \psi'(u_1(x))u_1'(x)$$

and

$$u_2''(x) = \psi'(u_1(x))u_1''(x) + \psi''(u_1(x))(u_1'(x))^2.$$

Dividing both sides of the second expression by $u_2'(x) > 0$, and using the first

19. In other words, any risk that $u_2(\cdot)$ would accept starting from a position of certainty would also be accepted by $u_1(\cdot)$.

expression, we get

$$r_A(x, u_2) = r_A(x, u_1) - \frac{\psi''(u_1(x))}{\psi'(u_1(x))} u_1'(x).$$

Thus, $r_A(x, u_2) \geq r_A(x, u_1)$ for all x if and only if $\psi''(u_1) \leq 0$ for all u_1 in the range of $u_1(\cdot)$. ■

The more-risk-averse-than relation is a *partial ordering* of Bernoulli utility functions; it is transitive but far from complete. Typically, two Bernoulli utility functions $u_1(\cdot)$ and $u_2(\cdot)$ will not be comparable; that is, we will have $r_A(x, u_1) > r_A(x, u_2)$ at some x but $r_A(x', u_1) < r_A(x', u_2)$ at some other $x' \neq x$.

Example 6.C.2 continued: We take up again the asset portfolio problem between a safe and a risky asset discussed in Example 6.C.2. Suppose that we now have two individuals with Bernoulli utility functions $u_1(\cdot)$ and $u_2(\cdot)$, and denote by α_1^* and α_2^* their respective optimal investments in the risky asset. We will show that if $u_2(\cdot)$ is more risk averse than $u_1(\cdot)$, then $\alpha_2^* < \alpha_1^*$; that is, the second decision maker invests less in the risky asset than the first.

To repeat from our earlier discussion, the asset allocation problem for $u_1(\cdot)$ is

$$\text{Max}_{0 \leq \alpha \leq w} \int u_1(w - \alpha + \alpha z) dF(z).$$

Assuming an interior solution, the first-order condition is

$$\int (z - 1) u_1'(w + \alpha_1^*[z - 1]) dF(z) = 0. \quad (6.C.5)$$

The analogous expression for the utility function $u_2(\cdot)$ is

$$\phi_2(\alpha_2^*) = \int (z - 1) u_2'(w + \alpha_2^*[z - 1]) dF(z) = 0. \quad (6.C.6)$$

As we know, the concavity of $u_2(\cdot)$ implies that $\phi_2(\cdot)$ is decreasing. Therefore, if we show that $\phi_2(\alpha_1^*) < 0$, it must follow that $\alpha_2^* < \alpha_1^*$, which is the result we want. Now, $u_2(x) = \psi(u_1(x))$ allows us to write

$$\phi_2(\alpha_1^*) = \int (z - 1) \psi'(u_1(w + \alpha_1^*[z - 1])) u_1'(w + \alpha_1^*[z - 1]) dF(z) < 0. \quad (6.C.7)$$

To understand the final inequality, note that the integrand of expression (6.C.7) is the same as that in (6.C.5) except that it is multiplied by $\psi'(\cdot)$, a positive decreasing function of z [recall that $u_2(\cdot)$ more risk averse than $u_1(\cdot)$ means that the increasing function $\psi(\cdot)$ is concave; that is, $\psi'(\cdot)$ is positive and decreasing]. Hence, the integral (6.C.7) underweights the positive values of $(z - 1) u_1'(w + \alpha_1^*[z - 1])$, which obtain for $z > 1$, relative to the negative values, which obtain for $z < 1$. Since, in (6.C.5), the integral of the positive and the negative parts of the integrand added to zero, they now must add to a negative number. This establishes the desired inequality. ■

Comparisons across wealth levels

It is a common contention that wealthier people are willing to bear more risk than poorer people. Although this might be due to differences in utility functions across people, it is more likely that the source of the difference lies in the possibility that

richer people “can afford to take a chance.” Hence, we shall explore the implications of the condition stated in Definition 6.C.4.

Definition 6.C.4: The Bernoulli utility function $u(\cdot)$ for money exhibits *decreasing absolute risk aversion* if $r_A(x, u)$ is a decreasing function of x .

Individuals whose preferences satisfy the decreasing absolute risk aversion property take more risk as they become wealthier. Consider two levels of initial wealth $x_1 > x_2$. Denote the increments or decrements to wealth by z . Then the individual evaluates risk at x_1 and x_2 by, respectively, the induced Bernoulli utility functions $u_1(z) = u(x_1 + z)$ and $u_2(z) = u(x_2 + z)$. Comparing an individual's attitudes toward risk as his level of wealth changes is like comparing the utility functions $u_1(\cdot)$ and $u_2(\cdot)$, a problem we have just studied. If $u(\cdot)$ displays decreasing absolute risk aversion, then $r_A(z, u_2) \geq r_A(z, u_1)$ for all z . This is condition (i) of Proposition 6.C.2. Hence, the result in Proposition 6.C.3 follows directly from Proposition 6.C.2.

Proposition 6.C.3: The following properties are equivalent:

- (i) The Bernoulli utility function $u(\cdot)$ exhibits decreasing absolute risk aversion.
- (ii) Whenever $x_2 < x_1$, $u_2(z) = u(x_2 + z)$ is a concave transformation of $u_1(z) = u(x_1 + z)$.
- (iii) For any risk $F(z)$, the certainty equivalent of the lottery formed by adding risk z to wealth level x , given by the amount c_x at which $u(c_x) = \int u(x + z) dF(z)$, is such that $(x - c_x)$ is decreasing in x . That is, the higher x is, the less is the individual willing to pay to get rid of the risk.
- (iv) The probability premium $\pi(x, \varepsilon, u)$ is decreasing in x .
- (v) For any $F(z)$, if $\int u(x_2 + z) dF(z) \geq u(x_2)$ and $x_2 < x_1$, then $\int u(x_1 + z) dF(z) \geq u(x_1)$.

Exercise 6.C.8: Assume that the Bernoulli utility function $u(\cdot)$ exhibits decreasing absolute risk aversion. Show that for the asset demand model of Example 6.C.2 (and Example 6.C.2 continued), the optimal allocation between the safe and the risky assets places an increasing amount of wealth in the risky asset as w rises (i.e., the risky asset is a normal good).

The assumption of decreasing absolute risk aversion yields many other economically reasonable results concerning risk-bearing behavior. However, in applications, it is often too weak and, because of its analytical convenience, it is sometimes complemented by a stronger assumption: *nonincreasing relative risk aversion*.

To understand the concept of relative risk aversion, note that the concept of absolute risk aversion is suited to the comparison of attitudes toward risky projects whose outcomes are *absolute gains or losses* from current wealth. But it is also of interest to evaluate risky projects whose outcomes are *percentage* gains or losses of current wealth. The concept of relative risk aversion does just this.

Let $t > 0$ stand for *proportional* increments or decrements of wealth. Then, an individual with Bernoulli utility function $u(\cdot)$ and initial wealth x can evaluate a random percentage risk by means of the utility function $\tilde{u}(t) = u(tx)$. The initial wealth position corresponds to $t = 1$. We already know that for a small risk around $t = 1$, the degree of risk aversion is well captured by $\tilde{u}''(1)/\tilde{u}'(1)$. Noting that $\tilde{u}''(1)/\tilde{u}'(1) = xu''(x)/u'(x)$, we are led to the concept stated in Definition 6.C.5.

Definition 6.C.5: Given a Bernoulli utility function $u(\cdot)$, the *coefficient of relative risk aversion* at x is $r_R(x, u) = -xu''(x)/u'(x)$.

Consider now how this measure varies with wealth. The property of *nonincreasing relative risk aversion* says that the individual becomes less risk averse with regard to gambles that are proportional to his wealth as his wealth increases. This is a stronger assumption than decreasing absolute risk aversion: Since $r_R(x, u) = xr_A(x, u)$, a risk-averse individual with decreasing relative risk aversion will exhibit decreasing absolute risk aversion, but the converse is not necessarily the case.

As before, we can examine various implications of this concept. Proposition 6.C.4 is an abbreviated parallel to Proposition 6.C.3.

Proposition 6.C.4: The following conditions for a Bernoulli utility function $u(\cdot)$ on amounts of money are equivalent:

- (i) $r_R(x, u)$ is decreasing in x .
- (ii) Whenever $x_2 < x_1$, $\tilde{u}_2(t) = u(tx_2)$ is a concave transformation of $\tilde{u}_1(t) = u(tx_1)$.
- (iii) Given any risk $F(t)$ on $t > 0$, the certainty equivalent $\bar{c}_x$ defined by $u(\bar{c}_x) = \int u(tx) dF(t)$ is such that $x/\bar{c}_x$ is decreasing in x .

Proof: Here we show only that (i) implies (iii). To this effect, fix a distribution $F(t)$ on $t > 0$, and, for any x , define $u_x(t) = u(tx)$. Let $c(x)$ be the usual certainty equivalent (from Definition 6.C.2): $u_x(c(x)) = \int u_x(t) dF(t)$. Note that $-u_x''(t)/u_x'(t) = -(1/t)tx[u''(tx)/u'(tx)]$ for any x . Hence if (i) holds, then $u_{x'}(\cdot)$ is less risk averse than $u_x(\cdot)$ whenever $x' > x$. Therefore, by Proposition 6.C.2, $c(x') > c(x)$ and we conclude that $c(\cdot)$ is increasing. Now, by the definition of $u_x(\cdot)$, $u_x(c(x)) = u(xc(x))$. Also

$$u_x(c(x)) = \int u_x(t) dF(t) = \int u(tx) dF(t) = u(\bar{c}_x).$$

Hence, $\bar{c}_x/x = c(x)$, and so $x/\bar{c}_x$ is decreasing. This concludes the proof. ■

Example 6.C.2 continued: In Exercise 6.C.11, you are asked to show that if $r_R(x, u)$ is decreasing in x , then the proportion of wealth invested in the risky asset $\gamma = \alpha/w$ is increasing with the individual's wealth level w . The opposite conclusion holds if $r_R(x, u)$ is increasing in x . If $r_R(x, u)$ is a constant independent of x , then the fraction of wealth invested in the risky asset is independent of w [see Exercise 6.C.12 for the specific analytical form that $u(\cdot)$ must have]. Models with constant relative risk aversion are encountered often in finance theory, where they lead to considerable analytical simplicity. Under this assumption, no matter how the wealth of the economy and its distribution across individuals evolves over time, the portfolio decisions of individuals in terms of budget shares do not vary (as long as the safe return and the distribution of random returns remain unchanged). ■

6.D Comparison of Payoff Distributions in Terms of Return and Risk

In this section, we continue our study of lotteries with monetary payoffs. In contrast with Section 6.C, where we compared utility functions, our aim here is to compare

payoff distributions. There are two natural ways that random outcomes can be compared: according to the level of returns and according to the dispersion of returns. We will therefore attempt to give meaning to two ideas: that of a distribution $F(\cdot)$ yielding unambiguously higher returns than $G(\cdot)$ and that of $F(\cdot)$ being unambiguously less risky than $G(\cdot)$. These ideas are known, respectively, by the technical terms of *first-order stochastic dominance* and *second-order stochastic dominance*.²⁰

In all subsequent developments, we restrict ourselves to distributions $F(\cdot)$ such that $F(0) = 0$ and $F(x) = 1$ for some x .

First-Order Stochastic Dominance

We want to attach meaning to the expression: "The distribution $F(\cdot)$ yields unambiguously higher returns than the distribution $G(\cdot)$." At least two sensible criteria suggest themselves. First, we could test whether every expected utility maximizer who values more over less prefers $F(\cdot)$ to $G(\cdot)$. Alternatively, we could verify whether, for every amount of money x , the probability of getting at least x is higher under $F(\cdot)$ than under $G(\cdot)$. Fortunately, these two criteria lead to the same concept.

Definition 6.D.1: The distribution $F(\cdot)$ *first-order stochastically dominates* $G(\cdot)$ if, for every nondecreasing function $u: \mathbb{R} \rightarrow \mathbb{R}$, we have

$$\int u(x) dF(x) \geq \int u(x) dG(x).$$

Proposition 6.D.1: The distribution of monetary payoffs $F(\cdot)$ first-order stochastically dominates the distribution $G(\cdot)$ if and only if $F(x) \leq G(x)$ for every x .

Proof: Given $F(\cdot)$ and $G(\cdot)$ denote $H(x) = F(x) - G(x)$. Suppose that $H(\bar{x}) > 0$ for some $\bar{x}$. Then we can define a nondecreasing function $u(\cdot)$ by $u(x) = 1$ for $x > \bar{x}$ and $u(x) = 0$ for $x \leq \bar{x}$. This function has the property that $\int u(x) dH(x) = -H(\bar{x}) < 0$, and so the "only if" part of the proposition follows.

For the "if" part of the proposition we first put on record, without proof, that it suffices to establish the equivalence for differentiable utility functions $u(\cdot)$. Given $F(\cdot)$ and $G(\cdot)$, denote $H(x) = F(x) - G(x)$. Integrating by parts, we have

$$\int u(x) dH(x) = [u(x)H(x)]_0^\infty - \int u'(x)H(x) dx.$$

Since $H(0) = 0$ and $H(x) = 0$ for large x , the first term of this expression is zero. It follows that $\int u(x) dH(x) \geq 0$ [or, equivalently, $\int u(x) dF(x) - \int u(x) dG(x) \geq 0$] if and only if $\int u'(x)H(x) dx \leq 0$. Thus, if $H(x) \leq 0$ for all x and $u(\cdot)$ is increasing, then $\int u'(x)H(x) dx \leq 0$ and the "if" part of the proposition follows. ■

In Exercise 6.D.1 you are asked to verify Proposition 6.D.1 for the case of lotteries over three possible outcomes. In Figure 6.D.1, we represent two distributions $F(\cdot)$ and $G(\cdot)$. Distribution $F(\cdot)$ first-order stochastically dominates $G(\cdot)$ because the graph of $F(\cdot)$ is uniformly below the graph of $G(\cdot)$. Note two important points: First, first-order stochastic dominance does *not* imply that every possible return of the

20. They were introduced into economics in Rothschild and Stiglitz (1970).

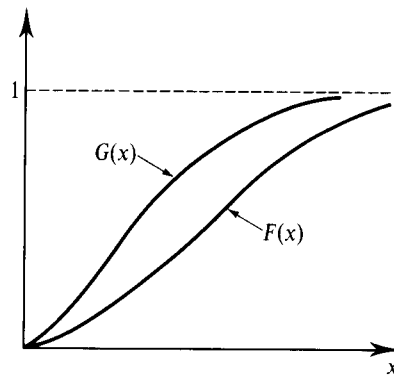


Figure 6.D.1
 $F(\cdot)$ first-order
 stochastically
 dominates $G(\cdot)$.

superior distribution is larger than every possible return of the inferior one. In the figure, the set of possible outcomes is the same for the two distributions. Second, although $F(\cdot)$ first-order stochastically dominates $G(\cdot)$ implies that the mean of x under $F(\cdot)$, $\int x dF(x)$, is greater than its mean under $G(\cdot)$, a ranking of the means of two distributions does *not* imply that one first-order stochastically dominates the other; rather, the entire distribution matters (see Exercise 6.D.3).

Example 6.D.1: Consider a compound lottery that has as its first stage a realization of x distributed according to $G(\cdot)$ and in its second stage applies to the outcome x of the first stage an “upward probabilistic shift.” That is, if outcome x is realized in the first stage, then the second stage pays a final amount of money $x + z$, where z is distributed according to a distribution $H_x(z)$ with $H_x(0) = 0$. Thus, $H_x(\cdot)$ generates a *final* return of at least x with probability one. (Note that the distributions applied to different x ’s may differ.)

Denote the resulting reduced distribution by $F(\cdot)$. Then for any nondecreasing function $u: \mathbb{R} \rightarrow \mathbb{R}$, we have

$$\int u(x) dF(x) = \int \left[\int u(x+z) dH_x(z) \right] dG(x) \geq \int u(x) dG(x).$$

So $F(\cdot)$ first-order stochastically dominates $G(\cdot)$.

A specific example is illustrated in Figure 6.D.2. As Figure 6.D.2(a) shows, $G(\cdot)$ is an even randomization between 1 and 4 dollars. The outcome “1 dollar” is then

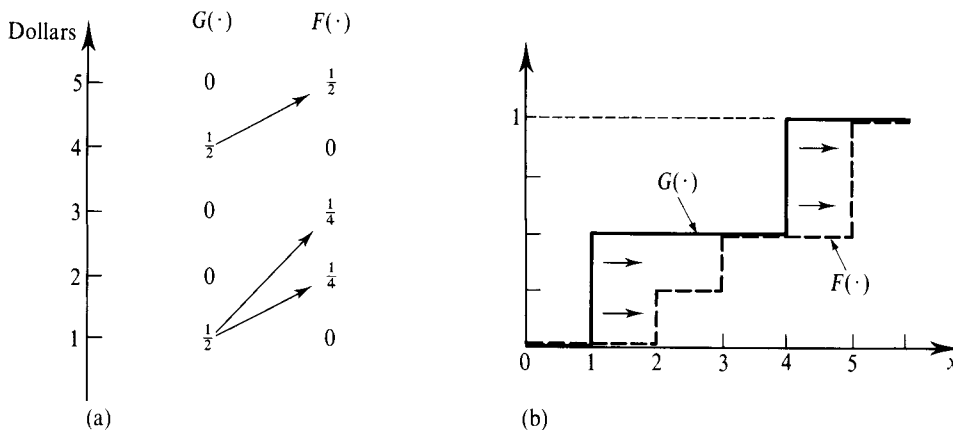


Figure 6.D.2
 $F(\cdot)$ first-order
 stochastically
 dominates $G(\cdot)$.

shifted up to an even probability between 2 and 3 dollars, and the outcome “4 dollars” is shifted up to 5 dollars with probability one. Figure 6.D.2(b) shows that $F(x) \leq G(x)$ at all x .

It can be shown that the reverse direction also holds. Whenever $F(\cdot)$ first-order stochastically dominates $G(\cdot)$, it is possible to generate $F(\cdot)$ from $G(\cdot)$ in the manner suggested in this example. Thus, this provides yet another approach to the characterization of the first-order stochastic dominance relation. ■

Second-Order Stochastic Dominance

First-order stochastic dominance involves the idea of “higher/better” vs. “lower/worse.” We want next to introduce a comparison based on relative *riskiness* or *dispersion*. To avoid confusing this issue with the trade-off between returns and risk, we will restrict ourselves for the rest of this section to comparing distributions with the same mean.

Once again, a definition suggests itself: Given two distributions $F(\cdot)$ and $G(\cdot)$ with the same mean [that is, with $\int x dF(x) = \int x dG(x)$], we say that $G(\cdot)$ is riskier than $F(\cdot)$ if every risk averter prefers $F(\cdot)$ and $G(\cdot)$. This is stated formally in Definition 6.D.2.

Definition 6.D.2: For any two distributions $F(x)$ and $G(\cdot)$ with the same mean, $F(\cdot)$ *second-order stochastically dominates* (or *is less risky than*) $G(\cdot)$ if for every nondecreasing concave function $u: \mathbb{R}_+ \rightarrow \mathbb{R}$, we have

$$\int u(x) dF(x) \geq \int u(x) dG(x).$$

Example 6.D.2 introduces an alternative way to characterize the second-order stochastic dominance relation.

Example 6.D.2: Mean-Preserving Spreads. Consider the following compound lottery: In the first stage, we have a lottery over x distributed according to $F(\cdot)$. In the second stage, we randomize each possible outcome x further so that the final payoff is $x + z$, where z has a distribution function $H_x(z)$ with a mean of zero [i.e., $\int z dH_x(z) = 0$]. Thus, the mean of $x + z$ is x . Let the resulting reduced lottery be denoted by $G(\cdot)$. When lottery $G(\cdot)$ can be obtained from lottery $F(\cdot)$ in this manner for some distribution $H_x(\cdot)$, we say that $G(\cdot)$ is a *mean-preserving spread* of $F(\cdot)$.

For example, $F(\cdot)$ may be an even probability distribution between 2 and 3 dollars. In the second step we may spread the 2 dollars outcome to an even probability between 1 and 3 dollars, and the 3 dollars outcome to an even probability between 2 and 4 dollars. Then $G(\cdot)$ is the distribution that assigns probability $\frac{1}{4}$ to the four outcomes: 1, 2, 3, 4 dollars. These two distributions $F(\cdot)$ and $G(\cdot)$ are depicted in Figure 6.D.3.

The type of two-stage operation just described keeps the mean of $G(\cdot)$ equal to that of $F(\cdot)$. In addition, if $u(\cdot)$ is concave, we can conclude that

$$\begin{aligned} \int u(x) dG(x) &= \int \left(\int u(x + z) dH_x(z) \right) dF(x) \leq \int u \left(\int (x + z) dH_x(z) \right) dF(x) \\ &= \int u(x) dF(x), \end{aligned}$$

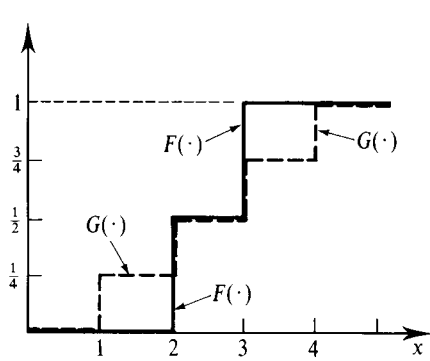


Figure 6.D.3 (left)
 $G(\cdot)$ is a mean-preserving spread of $F(\cdot)$.

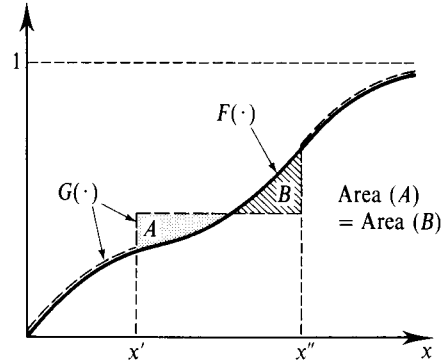


Figure 6.D.4 (right)
 $G(\cdot)$ is an elementary increase in risk from $F(\cdot)$.

and so $F(\cdot)$ second-order stochastically dominates $G(\cdot)$. It turns out that the converse is also true: If $F(\cdot)$ second-order stochastically dominates $G(\cdot)$, then $G(\cdot)$ is a mean-preserving spread of $F(\cdot)$. Hence, saying that $G(\cdot)$ is a mean-preserving spread of $F(\cdot)$ is equivalent to saying that $F(\cdot)$ second-order stochastically dominates $G(\cdot)$. ■

Example 6.D.3 provides another illustration of a mean-preserving spread.

Example 6.D.3: An Elementary Increase in Risk. We say that $G(\cdot)$ constitutes an elementary increase in risk from $F(\cdot)$ if $G(\cdot)$ is generated from $F(\cdot)$ by taking all the mass that $F(\cdot)$ assigns to an interval $[x', x'']$ and transferring it to the endpoints x' and x'' in such a manner that the mean is preserved. This is illustrated in Figure 6.D.4. An elementary increase in risk is a mean-preserving spread. [In Exercise 6.D.3, you are asked to verify directly that if $G(\cdot)$ is an elementary increase in risk from $F(\cdot)$, then $F(\cdot)$ second-order stochastically dominates $G(\cdot)$.] ■

We can develop still another way to capture the second-order stochastic dominance idea. Suppose that we have two distributions $F(\cdot)$ and $G(\cdot)$ with the same mean. Recall that, for simplicity, we assume that $F(\bar{x}) = G(\bar{x}) = 1$ for some $\bar{x}$. Integrating by parts (and recalling the equality of the means) yields

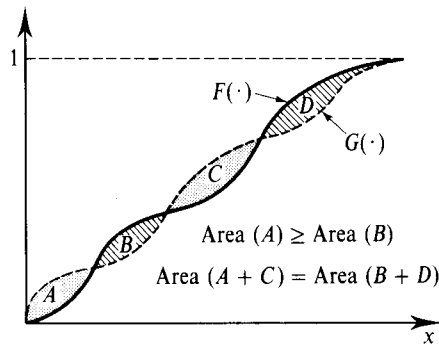
$$\int_0^{\bar{x}} (F(x) - G(x)) dx = - \int_0^{\bar{x}} x d(F(x) - G(x)) + (F(\bar{x}) - G(\bar{x}))\bar{x} = 0. \quad (6.D.1)$$

That is, the areas below the two distribution functions are the same over the interval $[0, \bar{x}]$. Because of this fact, the regions marked A and B in Figure 6.D.4 must have the same area. Note that for the two distributions in the figure, this implies that

$$\int_0^x G(t) dt \geq \int_0^x F(t) dt \quad \text{for all } x. \quad (6.D.2)$$

It turns out that property (6.D.2) is equivalent to $F(\cdot)$ second-order stochastically dominating $G(\cdot)$.²¹ As an application, suppose that $F(\cdot)$ and $G(\cdot)$ have the same mean and that the graph of $G(\cdot)$ is initially above the graph of $F(\cdot)$ and then moves

21. We will not prove this. The claim can be established along the same lines used to prove Proposition 6.D.1 except that we must integrate by parts twice and take into account expression (6.D.1).

**Figure 6.D.5**

$F(\cdot)$ second-order stochastically dominates $G(\cdot)$.

permanently below it (as in Figures 6.D.3 and 6.D.4). Then because of (6.D.1), condition (6.D.2) must be satisfied, and we can conclude that $G(\cdot)$ is riskier than $F(\cdot)$. As a more elaborate example, consider Figure 6.D.5, which shows two distributions having the same mean and satisfying (6.D.2). To verify that (6.D.2) is satisfied, note that area A has been drawn to be at least as large as area B and that the equality of the means [i.e., (6.D.1)] implies that the areas $B + D$ and $A + C$ must be equal.

We state Proposition 6.D.2 without proof.

Proposition 6.D.2: Consider two distributions $F(\cdot)$ and $G(\cdot)$ with the same mean. Then the following statements are equivalent:

- (i) $F(\cdot)$ second-order stochastically dominates $G(\cdot)$.
- (ii) $G(\cdot)$ is a mean-preserving spread of $F(\cdot)$.
- (iii) Property (6.D.2) holds.

In Exercise 6.D.4, you are asked to verify the equivalence of these three properties in the probability simplex diagram.

6.E State-dependent Utility

In this section, we consider an extension of the analysis presented in the preceding two sections. In Sections 6.C and 6.D, we assumed that the decision maker cares solely about the distribution of monetary payoffs he receives. This says, in essence, that the underlying cause of the payoff is of no importance. If the cause is one's state of health, however, this assumption is unlikely to be fulfilled.²² The distribution function of monetary payoffs is then not the appropriate object of individual choice. Here we consider the possibility that the decision maker may care not only about his monetary returns but also about the underlying events, or *states of nature*, that cause them.

We begin by discussing a convenient framework for modeling uncertain alternatives that, in contrast to the lottery apparatus, recognizes underlying states of nature. (We will encounter it repeatedly throughout the book, especially in Chapter 19.)

22. On the other hand, if it is an event such as the price of some security in a portfolio, the assumption is more likely to be a good representation of reality.

State-of-Nature Representations of Uncertainty

In Sections 6.C and 6.D, we modeled a risky alternative by means of a distribution function over monetary outcomes. Often, however, we know that the random outcome is generated by some underlying causes. A more detailed description of uncertain alternatives is then possible. For example, the monetary payoff of an insurance policy might depend on whether or not a certain accident has happened, the payoff on a corporate stock on whether the economy is in a recession, and the payoff of a casino gamble on the number selected by the roulette wheel.

We call these underlying causes *states*, or *states of nature*. We denote the set of states by S and an individual state by $s \in S$. For simplicity, we assume here that the set of states is finite and that each state s has a well-defined, objective probability $\pi_s > 0$ that it occurs. We abuse notation slightly by also denoting the total number of states by S .

An uncertain alternative with (nonnegative) monetary returns can then be described as a function that maps realizations of the underlying state of nature into the set of possible money payoffs $\mathbb{R}_+$. Formally, such a function is known as a *random variable*.

Definition 6.E.1: A *random variable* is a function $g: S \rightarrow \mathbb{R}_+$ that maps states into monetary outcomes.²³

Every random variable $g(\cdot)$ gives rise to a money lottery describable by the distribution function $F(\cdot)$ with $F(x) = \sum_{\{s: g(s) \leq x\}} \pi_s$ for all x . Note that there is a loss in information in going from the random variable representation of uncertainty to the lottery representation; we do not keep track of which states give rise to a given monetary outcome, and only the aggregate probability of every monetary outcome is retained.

Because we take S to be finite, we can represent a random variable with monetary payoffs by the vector $(x_1, \dots, x_S)$, where x_s is the nonnegative monetary payoff in state s . The set of all nonnegative random variables is then $\mathbb{R}_+^S$.

State-Dependent Preferences and the Extended Expected Utility Representation

The primitive datum of our theory is now a rational preference relation $\succeq$ on the set $\mathbb{R}_+^S$ of nonnegative random variables. Note that this formal setting is parallel to the one developed in Chapters 2 to 4 for consumer choice. The similarity is not merely superficial. If we define commodity s as the random variable that pays one dollar if and only if state s occurs (this is called a *contingent commodity* in Chapter 19), then the set of nonnegative random variables $\mathbb{R}_+^S$ is precisely the set of nonnegative bundles of these S contingent commodities.

As we shall see, it is very convenient if, in the spirit of the previous sections of this chapter, we can represent the individual's preferences over monetary outcomes by a utility function that possesses an *extended expected utility form*.

23. For concreteness, we restrict the outcomes to be nonnegative amounts of money. As we did in Section 6.B, we could equally well use an abstract outcome set C instead of $\mathbb{R}_+$.

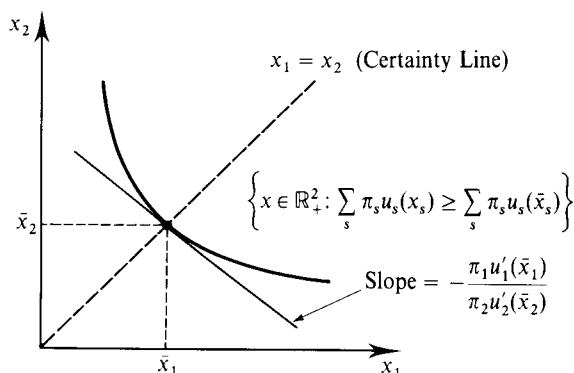


Figure 6.E.1
State-dependent preferences.

Definition 6.E.2: The preference relation $\succsim$ has an *extended expected utility representation* if for every $s \in S$, there is a function $u_s: \mathbb{R}_+ \rightarrow \mathbb{R}$ such that for any $(x_1, \dots, x_S) \in \mathbb{R}_+^S$ and $(x'_1, \dots, x'_S) \in \mathbb{R}_+^S$,

$$(x_1, \dots, x_S) \succsim (x'_1, \dots, x'_S) \quad \text{if and only if} \quad \sum_s \pi_s u_s(x_s) \geq \sum_s \pi_s u_s(x'_s).$$

To understand Definition 6.E.2, recall the analysis in Section 6.B. If only the distribution of money payoffs mattered, and if preferences on money distributions satisfied the expected utility axioms, then the expected utility theorem leads to a *state-independent* (we will also say *state-uniform*) expected utility representation $\sum_s \pi_s u(x_s)$, where $u(\cdot)$ is the Bernoulli utility function on amounts of money.²⁴ The generalization in Definition 6.E.2 allows for a different function $u_s(\cdot)$ in every state.

Before discussing the conditions under which an extended utility representation exists, we comment on its usefulness as a tool in the analysis of choice under uncertainty. This usefulness is primarily a result of the behavior of the indifference sets around the *money certainty line*, the set of random variables that pay the same amount in every state. Figure 6.E.1 depicts state-dependent preferences in the space $\mathbb{R}_+^S$ for a case where $S = 2$ and the $u_s(\cdot)$ functions are concave (as we shall see later, concavity of these functions follows from risk aversion considerations). The certainty line in Figure 6.E.1 is the set of points with $x_1 = x_2$. The marginal rate of substitution at a point $(\bar{x}, \bar{x})$ is $\pi_1 u'_1(\bar{x}) / \pi_2 u'_2(\bar{x})$. Thus, the slope of the indifference curves on the certainty line reflects the nature of state dependence as well as the probabilities of the different states. In contrast, with state-uniform (i.e., identical across states) utility functions, the marginal rate of substitution at any point on the certainty line equals the ratio of the probabilities of the states (implying that this slope is the same at all points on the certainty line).

Example 6.E.1: Insurance with State-dependent Utility. One interesting implication of state dependency arises when actuarially fair insurance is available. Suppose there are two states: State 1 is the state where no loss occurs, and state 2 is the state where a loss occurs. (This economic situation parallels that in Example 6.C.1.) The individual's initial situation (i.e., in the absence of any insurance purchase) is a

24. Note that the random variable $(x_1, \dots, x_S)$ induces a money lottery that pays x_s with probability π_s . Hence, $\sum_s \pi_s u(x_s)$ is its expected utility.

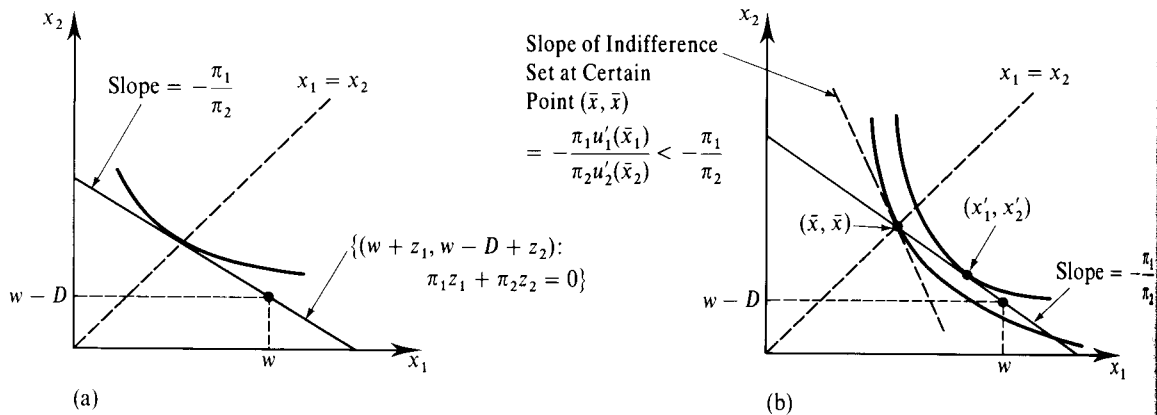


Figure 6.E.2 Insurance purchase with state-dependent utility. (a) State-uniform utility. (b) State-dependent utility.

random variable $(w, w - D)$ that gives the individual's wealth in the two states. This is depicted in Figure 6.E.2(a). We can represent an insurance contract by a random variable $(z_1, z_2) \in \mathbb{R}^2$ specifying the net change in wealth in the two states (the insurance payoff in the state less any premiums paid). Thus, if the individual purchases insurance contract (z_1, z_2) , his final wealth position will be $(w + z_1, w - D + z_2)$. The insurance policy (z_1, z_2) is actuarially fair if its expected payoff is zero, that is, if $\pi_1 z_1 + \pi_2 z_2 = 0$.

Figure 6.E.2(a) shows the optimal insurance purchase when a risk-averse expected utility maximizer with state-uniform preferences can purchase any actuarially fair insurance policy he desires. His budget set is the straight line drawn in the figure. We saw in Example 6.C.2 that under these conditions, a decision maker with state-uniform utility would insure completely. This is confirmed here because if there is no state dependency, the budget line is tangent to an indifference curve at the certainty line.

Figure 6.E.2(b) depicts the situation with state-dependent preferences. The decision maker will now prefer a point such as (x'_1, x'_2) to the certain outcome $(\bar{x}, \bar{x})$. This creates a desire to have a higher payoff in state 1, where $u'_1(\cdot)$ is relatively higher, in exchange for a lower payoff in state 2. ■

Existence of an Extended Expected Utility Representation

We now investigate conditions for the existence of an extended expected utility representation.

Observe first that since $\pi_s > 0$ for every s , we can formally include π_s in the definition of the utility function at state s . That is, to find an extended expected utility representation, it suffices that there be functions $u_s(\cdot)$ such that

$$(x_1, \dots, x_S) \succsim (x'_1, \dots, x'_S) \quad \text{if and only if} \quad \sum_s \pi_s u_s(x_s) \geq \sum_s \pi_s u_s(x'_s).$$

This is because if such functions $u_s(\cdot)$ exist, then we can define $\tilde{u}_s(\cdot) = (1/\pi_s)u_s(\cdot)$ for each $s \in S$, and we will have $\sum_s \pi_s u_s(x_s) \geq \sum_s \pi_s u_s(x'_s)$ if and only if $\sum_s \pi_s \tilde{u}_s(x_s) \geq \sum_s \pi_s \tilde{u}_s(x'_s)$. Thus, from now on, we focus on the existence of an additively separable form $\sum_s u_s(\cdot)$, and the π_s 's cease to play any role in the analysis.

It turns out that the extended expected utility representation can be derived in exactly the same way as the expected utility representation of Section 6.B if we appropriately enlarge the domain over which preferences are defined.²⁵ Accordingly, we now allow for the possibility that within each state s , the monetary payoff is not a certain amount of money x_s but a random amount with distribution function $F_s(\cdot)$. We denote these uncertain alternatives by $L = (F_1, \dots, F_S)$. Thus, L is a kind of compound lottery that assigns well-defined monetary gambles as prizes contingent on the realization of the state of the world s . We denote by $\mathcal{L}$ the set of all such possible lotteries.

Our starting point is now a rational preference relation $\succsim$ on $\mathcal{L}$. Note that $\alpha L + (1 - \alpha)L' = (\alpha F_1 + (1 - \alpha)F'_1, \dots, \alpha F_S + (1 - \alpha)F'_S)$ has the usual interpretation as the reduced lottery arising from a randomization between L and L' , although here we are dealing with a reduced lottery within each state s . Hence, we can appeal to the same logic as in Section 6.B and impose an independence axiom on preferences.

Definition 6.E.3: The preference relation $\succsim$ on $\mathcal{L}$ satisfies the *extended independence axiom* if for all $L, L', L'' \in \mathcal{L}$ and $\alpha \in (0, 1)$, we have

$$L \succsim L' \text{ if and only if } \alpha L + (1 - \alpha)L'' \succsim \alpha L' + (1 - \alpha)L''.$$

We also make a continuity assumption: Except for the reinterpretation of $\mathcal{L}$, this *continuity axiom* is exactly the same as that in Section 6.B; we refer to Definition 6.B.3 for its statement.

Proposition 6.E.1: (Extended Expected Utility Theorem) Suppose that the preference relation $\succsim$ on the space of lotteries $\mathcal{L}$ satisfies the continuity and extended independence axioms. Then we can assign a utility function $u_s(\cdot)$ for money in every state s such that for any $L = (F_1, \dots, F_S)$ and $L' = (F'_1, \dots, F'_S)$, we have

$$L \succsim L' \text{ if and only if } \sum_s \left(\int u_s(x_s) dF_s(x_s) \right) \geq \sum_s \left(\int u_s(x_s) dF'_s(x_s) \right).$$

Proof: The proof is identical, almost word for word, to the proof of the expected utility theorem (Proposition 6.B.2).

Suppose, for simplicity, that we restrict ourselves to a finite number $\{x_1, \dots, x_N\}$ of monetary outcomes. Then we can identify the set $\mathcal{L}$ with Δ^S , where Δ is the $(N - 1)$ -dimensional simplex. Our aim is to show that $\succsim$ can be represented by a linear utility function $U(L)$ on Δ^S . To see this, note that, up to an additive constant that can be neglected, $U(p_1^1, \dots, p_N^1, \dots, p_1^S, \dots, p_N^S)$ is a linear function of its arguments if it can be written as $U(L) = \sum_{n,s} u_{n,s} p_n^s$ for some values $u_{n,s}$. In this case, we can write $U(L) = \sum_s (\sum_n u_{n,s} p_n^s)$, which, letting $u_s(x_n) = u_{n,s}$, is precisely the form of a utility function on $\mathcal{L}$ that we want.

Choose $\bar{L}$ and $\underline{L}$ such that $\bar{L} \succsim L \succsim \underline{L}$ for all $L \in \mathcal{L}$. As in the proof of Proposition 6.B.2, we can then define $U(L)$ by the condition

$$L \sim U(L)\bar{L} + (1 - U(L))\underline{L}.$$

Applying the extended independence axiom in exactly the same way as we applied the independence axiom in the proof of Proposition 6.B.2 yields the result that $U(L)$ is indeed a linear utility function on $\mathcal{L}$. ■

25. By pushing the enlargement further than we do here, it would even be possible to view the existence of an extended utility representation as a corollary of the expected utility theorem.

Proposition 6.F.1 gives us a utility representation $\sum_s u_s(x_s)$ for the preferences on state-by-state sure outcomes $(x_1, \dots, x_S) \in \mathbb{R}_+^S$ that has two properties. First, it is additively separable across states. Second, every $u_s(\cdot)$ is a Bernoulli utility function that can be used to evaluate lotteries over money payoffs in state s by means of expected utility. It is because of the second property that risk aversion (defined in exactly the same manner as in Section 6.C) is equivalent to the concavity of each $u_s(\cdot)$.

There is another approach to the extended expected utility representation that rests with the preferences $\succsim$ defined on $\mathbb{R}_+^S$ and does not appeal to preferences defined on a larger space. It is based on the so-called *sure-thing axiom*.

Definition 6.E.4: The preference relation $\succsim$ satisfies the *sure-thing axiom* if, for any subset of states $E \subset S$ (E is called an *event*), whenever $(x_1, \dots, x_S)$ and $(x'_1, \dots, x'_S)$ differ only in the entries corresponding to E (so that $x'_s = x_s$ for $s \notin E$), the preference ordering between $(x_1, \dots, x_S)$ and $(x'_1, \dots, x'_S)$ is independent of the particular (common) payoffs for states not in E . Formally, suppose that $(x_1, \dots, x_S)$, $(x'_1, \dots, x'_S)$, $(\bar{x}_1, \dots, \bar{x}_S)$, and $(\bar{x}'_1, \dots, \bar{x}'_S)$ are such that

$$\begin{aligned} \text{For all } s \notin E: \quad & x_s = x'_s \quad \text{and} \quad \bar{x}_s = \bar{x}'_s. \\ \text{For all } s \in E: \quad & x_s = \bar{x}_s \quad \text{and} \quad x'_s = \bar{x}'_s. \end{aligned}$$

Then $(\bar{x}_1, \dots, \bar{x}_S) \succsim (\bar{x}'_1, \dots, \bar{x}'_S)$ if and only if $(x_1, \dots, x_S) \succsim (x'_1, \dots, x'_S)$.

The intuitive content of this axiom is similar to that of the independence axiom. It simply says that if two random variables cannot be distinguished in the complement of E , then the ordering among them can depend only on the values they take on E . In other words, tastes conditional on an event should not depend on what the payoffs would have been in states that have not occurred.

If $\succsim$ admits an extended expected utility representation, the sure-thing axiom holds because then $(x_1, \dots, x_S) \succsim (x'_1, \dots, x'_S)$ if and only if $\sum_s (u_s(x_s) - u_s(x'_s)) \geq 0$, and any term of the sum with $x_s = x'_s$ will cancel. In the other direction we have Proposition 6.E.2.

Proposition 6.E.2: Suppose that there are at least three states and that the preferences $\succsim$ on $\mathbb{R}_+^S$ are continuous and satisfy the sure-thing axiom. Then $\succsim$ admits an extended expected utility representation.

Idea of Proof: A complete proof is too advanced to be given in any detail. One wants to show that under the assumptions, preferences admit an additively separable utility representation $\sum_s u_s(x_s)$. This is not easy to show, and it is not a result particularly related to uncertainty. The conditions for the existence of an additively separable utility function for continuous preferences on the positive orthant of a Euclidean space (i.e., the context of Chapter 3) are well understood; as it turns out, they are *formally identical* to the sure-thing axiom (see Exercise 3.G.4). ■

Although the sure-thing axiom yields an extended expected utility representation $\sum_s \pi_s u_s(x_s)$, we would emphasize that randomizations over monetary payoffs in a state s have not been considered in this approach, and therefore we cannot bring the idea of risk aversion to bear on the determination of the properties of $u_s(\cdot)$. Thus, the approach via the extended independence axiom assumes a stronger basic framework (preferences are defined on the set $\mathcal{L}$ rather than on the smaller $\mathbb{R}_+^S$), but it also yields stronger conclusions.

6.F Subjective Probability Theory

Up to this point in the development of the theory, we have been assuming that risk, summarized by means of numerical probabilities, is regarded as an objective fact by the decision maker. But this is rarely true in reality. Individuals make judgments about the chances of uncertain events that are not necessarily expressible in quantitative form. Even when probabilities are mentioned, as sometimes happens when a doctor discusses the likelihood of various outcomes of medical treatment, they are often acknowledged as imprecise *subjective* estimates.

It would be very helpful, both theoretically and practically, if we could assert that choices are made *as if* individuals held probabilistic beliefs. Even better, we would like that well-defined probabilistic beliefs be revealed by choice behavior. This is the intent of *subjective probability theory*. The theory argues that even if states of the world are not associated with recognizable, objective probabilities, consistency-like restrictions on preferences among gambles still imply that decision makers behave *as if* utilities were assigned to outcomes, probabilities were attached to states of nature, and decisions were made by taking expected utilities. Moreover, this rationalization of the decision maker's behavior with an expected utility function can be done uniquely (up to a positive linear transformation for the utility functions). The theory is therefore a far-reaching generalization of expected utility theory. The classical reference for subjective probability theory is Savage (1954), which is very readable but also advanced. It is, however, possible to gain considerable insight into the theory if one is willing to let the analysis be aided by the use of lotteries with objective random outcomes. This is the approach suggested by Anscombe and Aumann (1963), and we will follow it here.

We begin, as in Section 6.E, with a set of states $\{1, \dots, S\}$. The probabilities on $\{1, \dots, S\}$ are not given. In effect, we aim to *deduce* them. As before, a random variable with monetary payoffs is a vector $x = (x_1, \dots, x_S) \in \mathbb{R}_+^S$.²⁶ We also want to allow for the possibility that the monetary payoffs in a state are not certain but are themselves money lotteries with objective distributions F_s . Thus, our set of risky alternatives, denoted $\mathcal{L}$, is the set of all S -tuples $(F_1, \dots, F_S)$ of distribution functions.

Suppose now that we are given a rational preference relation $\succsim$ on $\mathcal{L}$. We assume that $\succsim$ satisfies the continuity and the extended independence axioms introduced in Section 6.E. Then, by Proposition 6.E.1, we conclude that there are $u_s(\cdot)$ such that any $(x_1, \dots, x_S) \in \mathbb{R}_+^S$ can be evaluated by $\sum_s u_s(x_s)$. In addition, $u_s(\cdot)$ is a Bernoulli utility function for money lotteries in state s .

The existence of the $u_s(\cdot)$ functions does not yet allow us to identify subjective probabilities. Indeed, for any $(\pi_1, \dots, \pi_S) \gg 0$, we could define $\tilde{u}_s(\cdot) = (1/\pi_s)u_s(\cdot)$, and we could then evaluate $(x_1, \dots, x_S)$ by $\sum_s \pi_s \tilde{u}_s(x_s)$. What is needed is some way to disentangle utilities from probabilities.

Consider an example. Suppose that a gamble that gives one dollar in state 1 and none in state 2 is preferred to a gamble that gives one dollar in state 2 and none in state 1. Provided *there is no reason to think that the labels of the states have any*

26. To be specific, we consider monetary payoffs here. All the subsequent arguments, however, work with arbitrary sets of outcomes.

particular influence on the value of money, it is then natural to conclude that the decision maker regards state 2 as less likely than state 1.

This example suggests an additional postulate. Preferences over money lotteries within state s should be the same as those within any other state s' ; that is, risk attitudes towards money gambles should be the same across states. To formulate such a property, we define the state s preferences $\succsim_s$ on state s lotteries by

$$F_s \succsim_s F'_s \quad \text{if} \quad \int u_s(x_s) dF_s(x_s) \geq \int u_s(x_s) dF'_s(x_s).$$

Definition 6.F.1: The state preferences $(\succsim_1, \dots, \succsim_S)$ on state lotteries are *state uniform* if $\succsim_s = \succsim_{s'}$ for any s and s' .

With state uniformity, $u_s(\cdot)$ and $u_{s'}(\cdot)$ can differ only by an increasing linear transformation. Therefore, there is $u(\cdot)$ such that, for all $s = 1, \dots, S$,

$$u_s(\cdot) = \pi_s u(\cdot) + \beta_s$$

for some $\pi_s > 0$ and β_s . Moreover, because we still represent the same preferences if we divide all π_s and β_s by a common constant, we can normalize the π_s so that $\sum_s \pi_s = 1$. These π_s are going to be our subjective probabilities.

Proposition 6.F.1: (Subjective Expected Utility Theorem) Suppose that the preference relation $\succsim$ on $\mathcal{L}$ satisfies the continuity and extended independence axioms. Suppose, in addition, that the derived state preferences are state uniform. Then there are probabilities $(\pi_1, \dots, \pi_S) \gg 0$ and a utility function $u(\cdot)$ on amounts of money such that for any $(x_1, \dots, x_S)$ and $(x'_1, \dots, x'_S)$ we have

$$(x_1, \dots, x_S) \succsim (x'_1, \dots, x'_S) \quad \text{if and only if} \quad \sum_s \pi_s u(x_s) \geq \sum_s \pi_s u(x'_s).$$

Moreover, the probabilities are uniquely determined, and the utility function is unique up to origin and scale.

Proof: Existence has already been proven. You are asked to establish uniqueness in Exercise 6.F.1. ■

The practical advantages of the subjective expected utility representation are similar to those of the objective version, which we discussed in Section 6.B, and we will not repeat them here. A major virtue of the theory is that it gives a precise, quantifiable, and operational meaning to uncertainty. It is, indeed, most pleasant to be able to remain in the familiar realm of the probability calculus.

But there are also problems. The plausibility of the axioms cannot be completely dissociated from the complexity of the choice situations. The more complex these become, the more strained even seemingly innocent axioms are. For example, is the completeness axiom reasonable for preferences defined on huge sets of random variables? Or consider the implicit axiom (often those are the most treacherous) that the situation can actually be formalized as indicated by the model. This posits the ability to list all conceivable states of the world (or, at least, a sufficiently disaggregated version of this list). In summary, every difficulty so far raised against our model of the rational consumer (i.e., to transitivity, to completeness, to independence) will apply with increased force to the current model.

There are also difficulties specific to the nonobjective nature of probabilities. We devote Example 6.F.1 to this point.

Example 6.F.1: This example is a variation of the *Ellsberg paradox*.²⁷ There are two urns, denoted R and H. Each urn contains 100 balls. The balls are either white or black. Urn R contains 49 white balls and 51 black balls. Urn H contains an unspecified assortment of balls. A ball has been randomly picked from each urn. Call them the *R-ball* and the *H-ball*, respectively. The color of these balls has not been disclosed. Now we consider two choice situations. In both experiments, the decision maker must choose either the R-ball or the H-ball. After the choices have been made, the color will be disclosed. In the first choice situation, a prize of 1000 dollars is won if the chosen ball is black. In the second choice situation, the same prize is won if the ball is white. With the information given, most people will choose the R-ball in the first experiment. If the decision is made using subjective probabilities, this should mean that the subjective probability that the H-ball is white is larger than .49. Hence, most people should choose the H-ball in the second experiment. However, it turns out that this does not happen overwhelmingly in actual experiments. The decision maker understands that by choosing the R-ball, he has only a 49% chance of winning. However, this chance is “safe” and well understood. The uncertainties incurred are much less clear if he chooses the H-ball. ■

Knight (1921) proposed distinguishing between *risk* and *uncertainty* according to whether the probabilities are given to us objectively or not. In a sense, the theory of subjective probability nullifies this distinction by reducing all uncertainty to risk through the use of beliefs expressible as probabilities. The Example 6.F.1 suggests that there may be something to the distinction. This is an active area of research [e.g., Bewley (1986) and Gilboa and Schmeidler (1989)].

27. From Ellsberg (1961).

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EXERCISES

6.B.1^A In text.

6.B.2^A In text.

6.B.3^B Show that if the set of outcomes C is finite and the rational preference relation $\succeq$ on the set of lotteries $\mathcal{L}$ satisfies the independence axiom, then there are best and worst lotteries in $\mathcal{L}$. That is, we can find lotteries $\bar{L}$ and $\underline{L}$ such that $\bar{L} \succeq L \succeq \underline{L}$ for all $L \in \mathcal{L}$.

6.B.4^B The purpose of this exercise is to illustrate how expected utility theory allows us to make consistent decisions when dealing with extremely small probabilities by considering relatively large ones. Suppose that a safety agency is thinking of establishing a criterion under which an area prone to flooding should be evacuated. The probability of flooding is 1%. There are four possible outcomes:

- (A) No evacuation is necessary, and none is performed.
- (B) An evacuation is performed that is unnecessary.
- (C) An evacuation is performed that is necessary.
- (D) No evacuation is performed, and a flood causes a disaster.

Suppose that the agency is indifferent between the sure outcome B and the lottery of A with probability p and D with probability $1 - p$, and between the sure outcome C and the lottery of B with probability q and D with probability $1 - q$. Suppose also that it prefers A to D and that $p \in (0, 1)$ and $q \in (0, 1)$. Assume that the conditions of the expected utility theorem are satisfied.

- (a) Construct a utility function of the expected utility form for the agency.
- (b) Consider two different policy criteria:

Criterion 1: This criterion will result in an evacuation in 90% of the cases in which flooding will occur and an unnecessary evacuation in 10% of the cases in which no flooding occurs.

Criterion 2: This criterion is more conservative. It results in an evacuation in 95% of the cases in which flooding will occur and an unnecessary evacuation in 5% of the cases in which no flooding occurs.

First, derive the probability distributions over the four outcomes under these two criteria. Then, by using the utility function in (a), decide which criterion the agency would prefer.

6.B.5^B The purpose of this exercise is to show that the Allais paradox is compatible with a weaker version of the independence axiom. We consider the following axiom, known as the

betweenness axiom [see Dekel (1986)]:

For all L, L' and $\lambda \in (0, 1)$, if $L \sim L'$, then $\lambda L + (1 - \lambda)L' \sim L$.

Suppose that there are three possible outcomes.

(a) Show that a preference relation on lotteries satisfying the independence axiom also satisfies the betweenness axiom.

(b) Using a simplex representation for lotteries similar to the one in Figure 6.B.1(b), show that if the continuity and betweenness axioms are satisfied, then the indifference curves of a preference relation on lotteries are straight lines. Conversely, show that if the indifference curves are straight lines, then the betweenness axiom is satisfied. Do these straight lines need to be parallel?

(c) Using (b), show that the betweenness axiom is weaker (less restrictive) than the independence axiom.

(d) Using Figure 6.B.7, show that the choices of the Allais paradox are compatible with the betweenness axiom by exhibiting an indifference map satisfying the betweenness axiom that yields the choices of the Allais paradox.

6.B.6^B Prove that the induced utility function $U(\cdot)$ defined in the last paragraph of Section 6.B is convex. Give an example of a set of outcomes and a Bernoulli utility function for which the induced utility function is not linear.

6.B.7^A Consider the following two lotteries:

$$L: \begin{cases} 200 \text{ dollars with probability } .7. \\ 0 \text{ dollars with probability } .3. \end{cases}$$

$$L': \begin{cases} 1200 \text{ dollars with probability } .1. \\ 0 \text{ dollars with probability } .9. \end{cases}$$

Let x_L and $x_{L'}$ be the sure amounts of money that an individual finds indifferent to L and L' . Show that if his preferences are transitive and monotone, the individual must prefer L to L' if and only if $x_L > x_{L'}$. [Note: In actual experiments, however, a preference reversal is often observed in which L is preferred to L' but $x_L < x_{L'}$. See Grether and Plott (1979) for details.]

6.C.1^B Consider the insurance problem studied in Example 6.C.1. Show that if insurance is not actuarially fair (so that $q > \pi$), then the individual will not insure completely.

6.C.2^B

(a) Show that if an individual has a Bernoulli utility function $u(\cdot)$ with the quadratic form

$$u(x) = \beta x^2 + \gamma x,$$

then his utility from a distribution is determined by the mean and variance of the distribution and, in fact, by these moments alone. [Note: The number β should be taken to be negative in order to get the concavity of $u(\cdot)$. Since $u(\cdot)$ is then decreasing at $x > -\gamma/2\beta$, $u(\cdot)$ is useful only when the distribution cannot take values larger than $-\gamma/2\beta$.]

(b) Suppose that a utility function $U(\cdot)$ over distributions is given by

$$U(F) = (\text{mean of } F) - r(\text{variance of } F),$$

where $r > 0$. Argue that unless the set of possible distributions is further restricted (see, e.g., Exercise 6.C.19), $U(\cdot)$ cannot be compatible with any Bernoulli utility function. Give an example of two lotteries L and L' over the same two amounts of money, say x' and $x'' > x'$, such that L gives a higher probability to x'' than does L' and yet according to $U(\cdot)$, L' is preferred to L .

6.C.3^B Prove that the four conditions of Proposition 6.C.1 are equivalent. [Hint: The equivalence of (i), (ii), and (iii) has already been shown. As for (iv), prove that (i) implies (iv) and that (iv) implies $u(\frac{1}{2}x + \frac{1}{2}y) \geq \frac{1}{2}u(x) + \frac{1}{2}u(y)$ for any x and y , which is, in fact, sufficient for (ii).]

6.C.4^B Suppose that there are N risky assets whose returns z_n ($n = 1, \dots, N$) per dollar invested are jointly distributed according to the distribution function $F(z_1, \dots, z_N)$. Assume also that all the returns are nonnegative with probability one. Consider an individual who has a continuous, increasing, and concave Bernoulli utility function $u(\cdot)$ over $\mathbb{R}_+$. Define the utility function $U(\cdot)$ of this investor over $\mathbb{R}_+^N$, the set of all nonnegative portfolios, by

$$U(\alpha_1, \dots, \alpha_N) = \int u(\alpha_1 z_1 + \dots + \alpha_N z_N) dF(z_1, \dots, z_N).$$

Prove that $U(\cdot)$ is (a) increasing, (b) concave, and (c) continuous (this is harder).

6.C.5^A Consider a decision maker with utility function $u(\cdot)$ defined over $\mathbb{R}_+^L$, just as in Chapter 3.

(a) Argue that concavity of $u(\cdot)$ can be interpreted as the decision maker exhibiting risk aversion with respect to lotteries whose outcomes are bundles of the L commodities.

(b) Suppose now that a Bernoulli utility function $u(\cdot)$ for wealth is derived from the maximization of a utility function defined over bundles of commodities for each given wealth level w , while prices for those commodities are fixed. Show that, if the utility function for the commodities exhibits risk aversion, then so does the derived Bernoulli utility function for wealth. Interpret.

(c) Argue that the converse of part (b) does not need to hold: There are nonconcave functions $u: \mathbb{R}_+^L \rightarrow \mathbb{R}$ such that for any price vector the derived Bernoulli utility function on wealth exhibits risk aversion.

6.C.6^B For Proposition 6.C.2:

(a) Prove the equivalence of conditions (ii) and (iii).

(b) Prove the equivalence of conditions (iii) and (v).

6.C.7^A Prove that, in Proposition 6.C.2, condition (iii) implies condition (iv), and (iv) implies (i).

6.C.8^A In text.

6.C.9^B (M. Kimball) The purpose of this problem is to examine the implications of uncertainty and precaution in a simple consumption-savings decision problem.

In a two-period economy, a consumer has first-period initial wealth w . The consumer's utility level is given by

$$u(c_1, c_2) = u(c_1) + v(c_2),$$

where $u(\cdot)$ and $v(\cdot)$ are concave functions and c_1 and c_2 denote consumption levels in the first and the second period, respectively. Denote by x the amount saved by the consumer in the first period (so that $c_1 = w - x$ and $c_2 = x$), and let x_0 be the optimal value of x in this problem.

We now introduce uncertainty in this economy. If the consumer saves an amount x in the first period, his wealth in the second period is given by $x + y$, where y is distributed according to $F(\cdot)$. In what follows, $E[\cdot]$ always denotes the expectation with respect to $F(\cdot)$. Assume that the Bernoulli utility function over realized wealth levels in the two periods (w_1, w_2) is $u(w_1) + v(w_2)$. Hence, the consumer now solves

$$\text{Max}_x u(w - x) + E[v(x + y)].$$

Denote the solution to this problem by x^* .

(a) Show that if $E[v'(x_0 + y)] > v'(x_0)$, then $x^* > x_0$.

(b) Define the *coefficient of absolute prudence* of a utility function $v(\cdot)$ at wealth level x to be $-v'''(x)/v''(x)$. Show that if the coefficient of absolute prudence of a utility function $v_1(\cdot)$ is not larger than the coefficient of absolute prudence of utility function $v_2(\cdot)$ for all levels of wealth, then $E[v'_1(x_0 + y)] > v'_1(x_0)$ implies $E[v'_2(x_0 + y)] > v'_2(x_0)$. What are the implications of this fact in the context of part (a)?

(c) Show that if $v'''(\cdot) > 0$, and $E[y] = 0$, then $E[v'(x + y)] > v'(x)$ for all values of x .

(d) Show that if the coefficient of absolute risk aversion of $v(\cdot)$ is decreasing with wealth, then $-v'''(x)/v''(x) > -v''(x)/v'(x)$ for all x , and hence $v'''(\cdot) > 0$.

6.C.10^A Prove the equivalence of conditions (i) through (v) in Proposition 6.C.3. [Hint: By letting $u_1(z) = u(w_1 + z)$ and $u_2(z) = u(w_2 + z)$, show that each of the five conditions in Proposition 6.C.3 is equivalent to the counterpart in Proposition 6.C.2.]

6.C.11^B For the model in Example 6.C.2, show that if $r_R(x, u)$ is increasing in x then the proportion of wealth invested in the risky asset $\gamma = \alpha/x$ is decreasing with x . Similarly, if $r_R(x, u)$ is decreasing in x , then $\gamma = \alpha/x$ is increasing in x . [Hint: Let $u_1(t) = u(tw_1)$ and $u_2(t) = u(tw_2)$, and use the fact, stated in the analysis of Example 6.C.2, that if one Bernoulli utility function is more risk averse than another, then the optimal level of investment in the risky asset for the first function is smaller than that for the second function. You could also attempt a direct proof using first-order conditions.]

6.C.12^B Let $u: \mathbb{R}_+ \rightarrow \mathbb{R}$ be a strictly increasing Bernoulli utility function. Show that

(a) $u(\cdot)$ exhibits constant relative risk aversion equal to $\rho \neq 1$ if and only if $u(x) = \beta x^{1-\rho} + \gamma$, where $\beta > 0$ and $\gamma \in \mathbb{R}$.

(b) $u(\cdot)$ exhibits constant relative risk aversion equal to 1 if and only if $u(x) = \beta \ln x + \gamma$, where $\beta > 0$ and $\gamma \in \mathbb{R}$.

(c) $\lim_{\rho \rightarrow 1} (x^{1-\rho}/(1-\rho)) = \ln x$ for all $x > 0$.

6.C.13^B Assume that a firm is risk neutral with respect to profits and that if there is any uncertainty in prices, production decisions are made after the resolution of such uncertainty. Suppose that the firm faces a choice between two alternatives. In the first, prices are uncertain. In the second, prices are nonrandom and equal to the expected price vector in the first alternative. Show that a firm that maximizes expected profits will prefer the first alternative over the second.

6.C.14^B Consider two risk-averse decision makers (i.e., two decision makers with concave Bernoulli utility functions) choosing among monetary lotteries. Define the utility function $u^*(\cdot)$ to be strongly more risk averse than $u(\cdot)$ if and only if there is a positive constant k and a nonincreasing and concave function $v(\cdot)$ such that $u^*(x) = ku(x) + v(x)$ for all x . The monetary amounts are restricted to lie in the interval $[0, r]$.

(a) Show that if $u^*(\cdot)$ is strongly more risk averse than $u(\cdot)$, then $u^*(\cdot)$ is more risk averse than $u(\cdot)$ in the usual Arrow-Pratt sense.

(b) Show that if $u(\cdot)$ is bounded, then there is no $u^*(\cdot)$ other than $u^*(\cdot) = ku(\cdot) + c$, where c is a constant, that is strongly more risk averse than $u(\cdot)$ on the entire interval $[0, +\infty]$. [Hint: in this part, disregard the assumption that the monetary amounts are restricted to lie in the interval $[0, r]$.]

(c) Using (b), argue that the concept of a strongly more risk-averse utility function is stronger (i.e., more restrictive) than the Arrow-Pratt concept of a more risk-averse utility function.

6.C.15^A Assume that, in a world with uncertainty, there are two assets. The first is a riskless asset that pays 1 dollar. The second pays amounts a and b with probabilities of π and $1 - \pi$, respectively. Denote the demand for the two assets by (x_1, x_2) .

Suppose that a decision maker's preferences satisfy the axioms of expected utility theory and that he is a risk averter. The decision maker's wealth is 1, and so are the prices of the assets. Therefore, the decision maker's budget constraint is given by

$$x_1 + x_2 = 1, \quad x_1, x_2 \in [0, 1].$$

(a) Give a simple *necessary* condition (involving a and b only) for the demand for the riskless asset to be strictly positive.

(b) Give a simple *necessary* condition (involving a , b , and π only) for the demand for the risky asset to be strictly positive.

In the next three parts, assume that the conditions obtained in (a) and (b) are satisfied.

(c) Write down the first-order conditions for utility maximization in this asset demand problem.

(d) Assume that $a < 1$. Show by analyzing the first-order conditions that $dx_1/da \leq 0$.

(e) Which sign do you conjecture for $dx_1/d\pi$? Give an economic interpretation.

(f) Can you prove your conjecture in (e) by analyzing the first-order conditions?

6.C.16^A An individual has Bernoulli utility function $u(\cdot)$ and initial wealth w . Let lottery L offer a payoff of G with probability p and a payoff of B with probability $1 - p$.

(a) If the individual owns the lottery, what is the minimum price he would sell it for?

(b) If he does not own it, what is the maximum price he would be willing to pay for it?

(c) Are buying and selling prices equal? Give an economic interpretation for your answer. Find conditions on the parameters of the problem under which buying and selling prices are equal.

(d) Let $G = 10$, $B = 5$, $w = 10$, and $u(x) = \sqrt{x}$. Compute the buying and selling prices for this lottery and this utility function.

6.C.17^B Assume that an individual faces a two-period portfolio allocation problem. In period $t = 0, 1$, his wealth w_t is to be divided between a safe asset with return R and a risky asset with return x . The initial wealth at period 0 is w_0 . Wealth at period $t = 1, 2$ depends on the portfolio α_{t-1} chosen at period $t - 1$ and on the return x_t realized at period t , according to

$$w_t = ((1 - \alpha_{t-1})R + \alpha_{t-1}x_t)w_{t-1}.$$

The objective of this individual is to maximize the expected utility of terminal wealth w_2 . Assume that x_1 and x_2 are independently and identically distributed. Prove that the individual optimally sets $\alpha_0 = \alpha_1$ if his utility function exhibits constant relative risk aversion. Show also that this fails to hold if his utility function exhibits constant absolute risk aversion.

6.C.18^B Suppose that an individual has a Bernoulli utility function $u(x) = \sqrt{x}$.

(a) Calculate the Arrow-Pratt coefficients of absolute and relative risk aversion at the level of wealth $w = 5$.

(b) Calculate the certainty equivalent and the probability premium for a gamble $(16, 4; \frac{1}{2}, \frac{1}{2})$.

(c) Calculate the certainty equivalent and the probability premium for a gamble $(36, 16; \frac{1}{2}, \frac{1}{2})$. Compare this result with the one in (b) and interpret.

6.C.19^C Suppose that an individual has a Bernoulli utility function $u(x) = -e^{-\alpha x}$ where $\alpha > 0$. His (nonstochastic) initial wealth is given by w . There is one riskless asset and there are N

risky assets. The return per unit invested on the riskless asset is r . The returns of the risky assets are jointly normally distributed random variables with means $\mu = (\mu_1, \dots, \mu_N)$ and variance-covariance matrix V . Assume that there is no redundancy in the risky assets, so that V is of full rank. Derive the demand function for these $N + 1$ assets.

6.C.20^A Consider a lottery over monetary outcomes that pays $x + \varepsilon$ with probability $\frac{1}{2}$ and $x - \varepsilon$ with probability $\frac{1}{2}$. Compute the second derivative of this lottery's certainty equivalent with respect to ε . Show that the limit of this derivative as $\varepsilon \rightarrow 0$ is exactly $-r_A(x)$.

6.D.1^A The purpose of this exercise is to prove Proposition 6.D.1 in a two-dimensional probability simplex. Suppose that there are three monetary outcomes: 1 dollar, 2 dollars, and 3 dollars. Consider the probability simplex of Figure 6.B.1(b).

(a) For a given lottery L over these outcomes, determine the region of the probability simplex in which lie the lotteries whose distributions first-order stochastically dominate the distribution of L .

(b) Given a lottery L , determine the region of the probability simplex in which lie the lotteries L' such that $F(x) \leq G(x)$ for every x , where $F(\cdot)$ is the distribution of L' and $G(\cdot)$ is the distribution of L . [Notice that we get the same region as in (a).]

6.D.2^A Prove that if $F(\cdot)$ first-order stochastically dominates $G(\cdot)$, then the mean of x under $F(\cdot)$, $\int x dF(x)$, exceeds that under $G(\cdot)$, $\int x dG(x)$. Also provide an example where $\int x dF(x) > \int x dG(x)$ but $F(\cdot)$ does not first-order stochastically dominate $G(\cdot)$.

6.D.3^A Verify that if a distribution $G(\cdot)$ is an elementary increase in risk from a distribution $F(\cdot)$, then $F(\cdot)$ second-order stochastically dominates $G(\cdot)$.

6.D.4^A The purpose of this exercise is to verify the equivalence of the three statements of Proposition 6.D.2 in a two-dimensional probability simplex. Suppose that there are three monetary outcomes: 1, 2, and 3 dollars. Consider the probability simplex in Figure 6.B.1(b).

(a) If two lotteries have the same mean, what are their positions relative to each other in the probability simplex.

(b) Given a lottery L , determine the region of the simplex in which lie the lotteries L' whose distributions are second-order stochastically dominated by the distribution of L .

(c) Given a lottery L , determine the region of the simplex in which lie the lotteries L' whose distributions are mean preserving spreads of L .

(d) Given a lottery L , determine the region of the simplex in which lie the lotteries L' for which condition (6.D.2) holds, where $F(\cdot)$ and $G(\cdot)$ are, respectively, the distributions of L and L' .

Notice that in (b), (c), and (d), you always have the same region.

6.E.1^B The purpose of this exercise is to show that preferences may not be transitive in the presence of regret. Let there be S states of the world, indexed by $s = 1, \dots, S$. Assume that state s occurs with probability π_s . Define the expected regret associated with lottery $x = (x_1, \dots, x_S)$ relative to lottery $x' = (x'_1, \dots, x'_S)$ by

$$\sum_{s=1}^S \pi_s h(\text{Max} \{0, x'_s - x_s\}),$$

where $h(\cdot)$ is a given increasing function. [We call $h(\cdot)$ the *regret valuation function*; it measures the regret the individual has after the state of nature is known.] We define x to be at least as good as x' in the presence of regret if and only if the expected regret associated with x relative to x' is not greater than the expected regret associated with x' relative to x .

Suppose that $S = 3$, $\pi_1 = \pi_2 = \pi_3 = \frac{1}{3}$, and $h(x) = \sqrt{x}$. Consider the following three lotteries:

$$x = (0, -2, 1),$$

$$x' = (0, 2, -2),$$

$$x'' = (2, -3, -1).$$

Show that the preference ordering over these three lotteries is not transitive.

6.E.2^A Assume that in a world with uncertainty there are two possible states of nature ($s = 1, 2$) and a single consumption good. There is a single decision maker whose preferences over lotteries satisfy the axioms of expected utility theory and who is a risk averter. For simplicity, we assume that utility is state-independent.

Two contingent commodities are available to the decision maker. The first (respectively, the second) pays one unit of the consumption good in state $s = 1$ (respectively $s = 2$) and zero otherwise. Denote the vector quantities of the two contingent commodities by (x_1, x_2) .

(a) Show that the preference relation of the decision maker on (x_1, x_2) is convex.

(b) Argue that the decision maker is also a risk averter when choosing between lotteries whose outcomes are vectors (x_1, x_2) .

(c) Show that the Walrasian demand functions for x_1 and x_2 are normal.

6.E.3^B Let $g: S \rightarrow \mathbb{R}_+$ be a random variable with mean $E(g) = 1$. For $\alpha \in (0, 1)$, define a new random variable $g^*: S \rightarrow \mathbb{R}_+$ by $g^*(s) = \alpha g(s) + (1 - \alpha)$. Note that $E(g^*) = 1$. Denote by $G(\cdot)$ and $G^*(\cdot)$ the distribution functions of $g(\cdot)$ and $g^*(\cdot)$, respectively. Show that $G^*(\cdot)$ second-order stochastically dominates $G(\cdot)$. Interpret.

6.F.1^B Prove that in the subjective expected utility theorem (Proposition 6.F.2), the obtained utility function $u(\cdot)$ on money is uniquely determined up to origin and scale. That is, if both $u(\cdot)$ and $\hat{u}(\cdot)$ satisfy the condition of the theorem, then there exist $\beta > 0$ and $\gamma \in \mathbb{R}$ such that $\hat{u}(x) = \beta u(x) + \gamma$ for all x . Prove also that the subjective probabilities are uniquely determined.

6.F.2^A The purpose of this exercise is to explain the outcomes of the experiments described in Example 6.F.1 by means of the theory of *nonunique prior beliefs* of Gilboa and Schmeidler (1989).

We consider a decision maker with a Bernoulli utility function $u(\cdot)$ defined on $\{0, 1000\}$. We normalize $u(\cdot)$ so that $u(0) = 0$ and $u(1000) = 1$.

The probabilistic belief that the decision maker might have on the color of the H-ball being white is a number $\pi \in [0, 1]$. We assume that the decision maker has, not a single belief but a set of beliefs given by a subset P of $[0, 1]$. The actions that he may take are denoted R or H with R meaning that he chooses the R-ball and H meaning that he chooses the H-ball.

As in Example 6.F.1, the decision maker is faced with two different choice situations. In choice situation W , he receives 1000 dollars if the ball chosen is white and 0 dollars otherwise. In choice situation B , he receives 1000 dollars if the ball chosen is black and 0 dollars otherwise.

For each of the two choice situations, define his utility function over the actions R and H in the following way:

For situation W , $U_W: \{R, H\} \rightarrow \mathbb{R}$ is defined by

$$U_W(R) = .49 \quad \text{and} \quad U_W(H) = \text{Min} \{ \pi : \pi \in P \}.$$

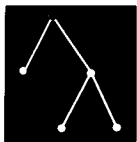
For situation B , $U_B: \{R, H\} \rightarrow \mathbb{R}$ is defined by

$$U_B(R) = .51 \quad \text{and} \quad U_B(H) = \text{Min} \{ (1 - \pi) : \pi \in P \}.$$

Namely, his utility from choice R is the expected utility of 1000 dollars with the (objective) probability calculated from the number of white and black balls in urn R. However, his utility from choice H is the expected utility of 1000 dollars with the probability associated with the most pessimistic belief in P .

(a) Prove that if P consists of only one belief, then U_W and U_B are derived from a von Neumann–Morgenstern utility function and that $U_W(R) > U_W(H)$ if and only if $U_B(R) < U_B(H)$.

(b) Find a set P for which $U_W(R) > U_W(H)$ and $U_B(R) > U_B(H)$.



Game Theory

In Part I, we analyzed individual decision making, both in abstract decision problems and in more specific economic settings. Our primary aim was to lay the groundwork for the study of how the simultaneous behavior of many self-interested individuals (including firms) generates economic outcomes in market economies. Most of the remainder of the book is devoted to this task. In Part II, however, we study in a more general way how multiperson interactions can be modeled.

A central feature of multiperson interaction is the potential for the presence of *strategic interdependence*. In our study of individual decision making in Part I, the decision maker faced situations in which her well-being depended only on the choices she made (possibly with some randomness). In contrast, in multiperson situations with strategic interdependence, each agent recognizes that the payoff she receives (in utility or profits) depends not only on her own actions but also on the actions of *other* individuals. The actions that are best for her to take may depend on actions these other individuals have already taken, on those she expects them to be taking at the same time, and even on future actions that they may take, or decide not to take, as a result of her current actions.

The tool that we use for analyzing settings with strategic interdependence is *noncooperative game theory*. Although the term “game” may seem to undersell the theory’s importance, it correctly highlights the theory’s central feature: The agents under study are concerned with strategy and winning (in the general sense of utility or profit maximization) in much the same way that players of most parlor games are.

Multiperson economic situations vary greatly in the degree to which strategic interaction is present. In settings of monopoly (where a good is sold by only a single firm; see Section 12.B) or of perfect competition (where all agents act as price takers; see Chapter 10 and Part IV), the nature of strategic interaction is minimal enough that our analysis need not make any formal use of game theory.¹ In other settings, however, such as the analysis of oligopolistic markets (where there is more than one

1. However, we could well do so in both cases; see, for example, the proof of existence of competitive equilibrium in Chapter 17, Appendix B. Moreover, we shall stress how perfect competition can be viewed usefully as a limiting case of oligopolistic strategic interaction; see, for example, Section 12.F.

but still not many sellers of a good; see Sections 12.C to 12.G), the central role of strategic interaction makes game theory indispensable for our analysis.

Part II is divided into three chapters. Chapter 7 provides a short introduction to the basic elements of noncooperative game theory, including a discussion of exactly what a game is, some ways of representing games, and an introduction to a central concept of the theory, a player's *strategy*. Chapter 8 addresses how we can predict outcomes in the special class of games in which all the players move simultaneously, known as *simultaneous-move games*. This restricted focus helps us isolate some central issues while deferring a number of more difficult ones. Chapter 9 studies *dynamic games* in which players' moves may precede one another, and in which some of these more difficult (but also interesting) issues arise.

Note that we have used the modifier *noncooperative* to describe the type of game theory we discuss in Part II. There is another branch of game theory, known as *cooperative game theory*, that we do not discuss here. In contrast with noncooperative game theory, the fundamental units of analysis in cooperative theory are groups and subgroups of individuals that are assumed, as a primitive of the theory, to be able to attain particular outcomes for themselves through binding cooperative agreements. Cooperative game theory has played an important role in general equilibrium theory, and we provide a brief introduction to it in Appendix A of Chapter 18. We should emphasize that the term *noncooperative game theory* does *not* mean that noncooperative theory is incapable of explaining cooperation within groups of individuals. Rather, it focuses on how cooperation may emerge as rational behavior in the absence of an ability to make binding agreements (e.g., see the discussion of repeated interaction among oligopolists in Chapter 12).

Some excellent recent references for further study of noncooperative game theory are Fudenberg and Tirole (1991), Myerson (1992), and Osborne and Rubinstein (1994), and at a more introductory level Gibbons (1992) and Binmore (1992). Kreps (1990) provides a very interesting discussion of some of the strengths and weaknesses of the theory. Von Neumann and Morgenstern (1944), Luce and Raiffa (1957), and Schelling (1960) remain classic references.

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Basic Elements of Noncooperative Games

7.A Introduction

In this chapter, we begin our study of noncooperative game theory by introducing some of its basic building blocks. This material serves as a prelude to our analysis of games in Chapters 8 and 9.

Section 7.B begins with an informal introduction to the concept of a *game*. It describes the four basic elements of any setting of strategic interaction that we must know to specify a game.

In Section 7.C, we show how a game can be described by means of what is called its *extensive form representation*. The extensive form representation provides a very rich description of a game, capturing who moves when, what they can do, what they know when it is their turn to move, and the outcomes associated with any collection of actions taken by the individuals playing the game.

In Section 7.D, we introduce a central concept of game theory, a player's *strategy*. A player's strategy is a complete contingent plan describing the actions she will take in each conceivable evolution of the game. We then show how the notion of a strategy can be used to derive a much more compact representation of a game, known as its *normal* (or *strategic*) *form representation*.

In Section 7.E, we consider the possibility that a player might randomize her choices. This gives rise to the notion of a *mixed strategy*.

7.B What Is a Game?

A *game* is a formal representation of a situation in which a number of individuals interact in a setting of *strategic interdependence*. By that, we mean that each individual's welfare depends not only on her own actions but also on the actions of the other individuals. Moreover, the actions that are best for her to take may depend on what she expects the other players to do.

To describe a situation of strategic interaction, we need to know four things:

- (i) *The players:* Who is involved?
- (ii) *The rules:* Who moves when? What do they know when they move?
What can they do?

- (iii) *The outcomes:* For each possible set of actions by the players, what is the outcome of the game?
- (iv) *The payoffs:* What are the players' preferences (i.e., utility functions) over the possible outcomes?

We begin by considering items (i) to (iii). A simple example is provided by the school-yard game of *Matching Pennies*.

Example 7.B.1: *Matching Pennies.* Items (i) to (iii) are as follows:

- Players:* There are two players, denoted 1 and 2.
- Rules:* Each player simultaneously puts a penny down, either heads up or tails up.
- Outcomes:* If the two pennies match (either both heads up or both tails up), player 1 pays 1 dollar to player 2; otherwise, player 2 pays 1 dollar to player 1. ■

Consider another example, the game of *Tick-Tack-Toe*.

Example 7.B.2: *Tick-Tack-Toe.* Items (i) to (iii) are as follows:

- Players:* There are two players, X and O.
- Rules:* The players are faced with a board that consists of nine squares arrayed with three rows of three squares each stacked on one another (see Figure 7.B.1). The players take turns putting their marks (an X or an O) into an as-yet-unmarked square. Player X moves first. Both players observe all choices previously made.
- Outcomes:* The first player to have three of her marks in a row (horizontally, vertically, or diagonally) wins and receives 1 dollar from the other player. If no one succeeds in doing so after all nine boxes are marked, the game is a tie and no payments are made or received by either player. ■

To complete our description of these two games, we need to say what the players' preferences are over the possible outcomes [item (iv) in our list]. As a general matter, we describe a player's preferences by a utility function that assigns a utility level for each possible outcome. It is common to refer to the player's utility function as her *payoff function* and the utility level as her *payoff*. Throughout, we assume that these utility functions take an expected utility form (see Chapter 6) so that when we consider situations in which outcomes are random, we can evaluate the random prospect by means of the player's expected utility.

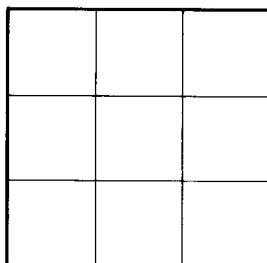


Figure 7.B.1
A Tick-Tack-Toe board.

In later references to Matching Pennies and Tick-Tack-Toe, we assume that each player's payoff is simply equal to the amount of money she gains or loses. Note that in both examples, the actions that maximize a player's payoff depend on what she expects her opponent to do.

Examples 7.B.1 and 7.B.2 involve situations of pure conflict: What one player wins, the other player loses. Such games are called *zero-sum games*. But strategic interaction and game theory are not limited to situations of pure or even partial conflict. Consider the situation in Example 7.B.3.

Example 7.B.3: *Meeting in New York.* Items (i) to (iv) are as follows:

- Players:* Two players, Mr. Thomas and Mr. Schelling.
- Rules:* The two players are separated and cannot communicate. They are supposed to meet in New York City at noon for lunch but have forgotten to specify where. Each must decide where to go (each can make only one choice).
- Outcomes:* If they meet each other, they get to enjoy each other's company at lunch. Otherwise, they must eat alone.
- Payoffs:* They each attach a monetary value of 100 dollars to the other's company (their payoffs are each 100 dollars if they meet, 0 dollars if they do not).

In this example, the two players' interests are completely aligned. Their problem is simply one of coordination. Nevertheless, each player's payoff depends on what the other player does; and more importantly, *each player's optimal action depends on what he thinks the other will do*. Thus, even the task of coordination can have a strategic nature. ■

Although the information given in items (i) to (iv) fully describe a game, it is useful for purposes of analysis to represent this information in particular ways. We examine one of these ways in Section 7.C.

7.C The Extensive Form Representation of a Game

If we know the items (i) to (iv) described in Section 7.B (the players, the rules, the outcomes, and the payoffs), then we can formally represent the game in what is called its *extensive form*. The extensive form captures who moves when, what actions each player can take, what players know when they move, what the outcome is as a function of the actions taken by the players, and the players' payoffs from each possible outcome.

We begin by informally introducing the elements of the extensive form representation through a series of examples. After doing so, we then provide a formal specification of the extensive form (some readers may want to begin with this and then return to the examples).

The extensive form relies on the conceptual apparatus known as a *game tree*. As our starting point, it is useful to begin with a very simple variation of Matching Pennies, which we call *Matching Pennies Version B*.

Example 7.C.1: *Matching Pennies Version B and Its Extensive Form.* Matching Pennies Version B is identical to Matching Pennies (see Example 7.B.1) except

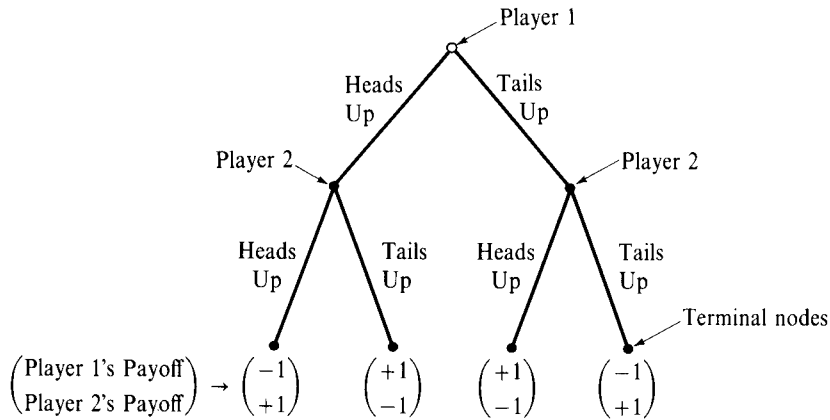


Figure 7.C.1
Extensive form for
Matching Pennies
Version B.

that the two players move sequentially, rather than simultaneously. In particular, player 1 puts her penny down (heads up or tails up) first. Then, after seeing player 1's choice, player 2 puts her penny down. (This is a very nice game for player 2!)

The extensive form representation of this game is depicted in Figure 7.C.1. The game starts at an *initial decision node* (represented by an open circle), where player 1 makes her move, deciding whether to place her penny heads up or tails up. Each of the two possible choices for player 1 is represented by a *branch* from this initial decision node. At the end of each branch is another decision node (represented by a solid dot), at which player 2 can choose between two actions, heads up or tails up, after seeing player 1's choice. The initial decision node is referred to as *player 1's decision node*; the latter two as *player 2's decision nodes*. After player 2's move, we reach the end of the game, represented by *terminal nodes*. At each terminal node, we list the players' payoffs arising from the sequence of moves leading to that terminal node.

Note the treelike structure of Figure 7.C.1: Like an actual tree, it has a unique connected path of branches from the initial node (sometimes also called the *root*) to each point in the tree. This type of figure is known as a *game tree*. ■

Example 7.C.2: The Extensive Form of Tick-Tack-Toe. The more elaborate game tree shown in Figure 7.C.2 depicts the extensive form for Tick-Tack-Toe (to conserve space, many parts are omitted). Note that every path through the tree represents a unique sequence of moves by the players. In particular, when a given board position (such as the two left corners filled by X and the two right corners filled by O) can be reached through several different sequences of moves, each of these sequences is depicted separately in the game tree. Nodes represent not only the current position but also *how it was reached*. ■

In both Matching Pennies Version B and Tick-Tack-Toe, when it is a player's turn to move, she is able to observe all her rival's previous moves. They are games of *perfect information* (we give a precise definition of this term in Definition 7.C.1). The concept of an *information set* allows us to accommodate the possibility that this is not so. Formally, the elements of an information set are a subset of a particular player's decision nodes. The interpretation is that when play has reached one of the decision nodes in the information set and it is that player's turn to move, she does

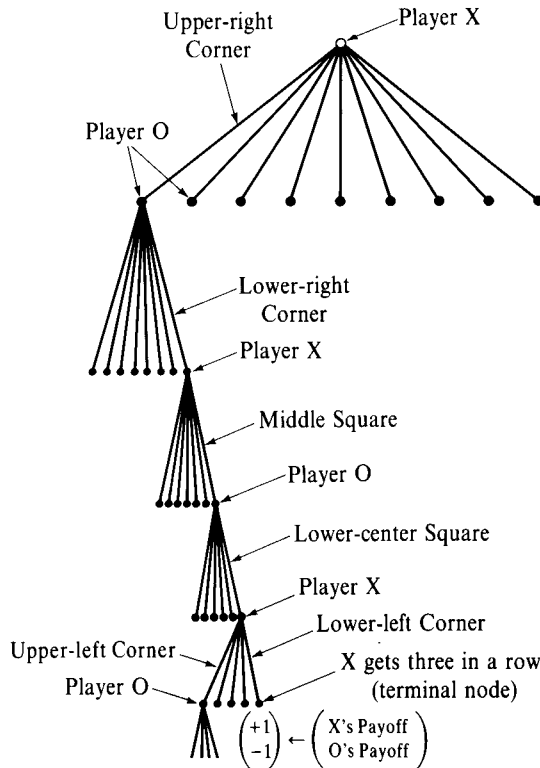


Figure 7.C.2
Part of the extensive
form for
Tick-Tack-Toe.

not know which of these nodes she is actually at. The reason for this ignorance is that the player does not observe something about what has previously transpired in the game. A further variation of Matching Pennies, which we call *Matching Pennies Version C*, helps make this concept clearer.

Example 7.C.3: Matching Pennies Version C and Its Extensive Form. This version of Matching Pennies is just like Matching Pennies Version B (in Example 7.C.1) except that when player 1 puts her penny down, she keeps it covered with her hand. Hence, player 2 cannot see player 1's choice until after player 2 has moved.

The extensive form for this game is represented in Figure 7.C.3. It is identical to Figure 7.C.1 except that we have drawn a circle around player 2's two decision nodes to indicate that these two nodes are in a single information set. The meaning of this information set is that when it is player 2's turn to move, she cannot tell which of these two nodes she is at because she has not observed player 1's previous move. Note that player 2 has the same two possible actions at each of the two nodes in her information set. This must be the case if player 2 is unable to distinguish the two nodes; otherwise, she could figure out which move player 1 had taken simply by what her own possible actions are.

In principle, we could also associate player 1's decision node with an information set. Because player 1 knows that nothing has happened before it is her turn to move, this information set has only one member (player 1 knows exactly which node she is at when she moves). To be fully rigorous, we should therefore also draw an information set circle around player 1's decision node in Figure 7.C.3. It is common, however, to

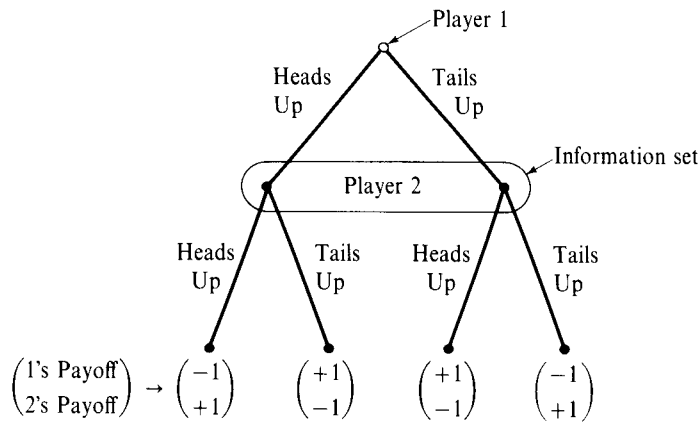


Figure 7.C.3
Extensive form for
Matching Pennies
Version C.

simplify the diagrammatic depiction of a game in extensive form by not drawing the information sets that contain a single node. Thus, any uncircled decision nodes are understood to be elements of *singleton* information sets. In Figures 7.C.1 and 7.C.2, for example, every decision node belongs to a singleton information set. ■

A listing of all of a player's information sets gives a listing, from the player's perspective, of all of the possible distinguishable "events" or "circumstances" in which she might be called upon to move. For example, in Example 7.C.1, from player 2's perspective there are two distinguishable events that might arise in which she would be called upon to move, each one corresponding to play having reached one of her two (singleton) information sets. By way of contrast, player 2 foresees only one possible circumstance in which she would need to move in Example 7.C.3 (this circumstance is, however, certain to arise).

In Example 7.C.3, we noted a natural restriction on information sets: At every node within a given information set, a player must have the same set of possible actions. Another restriction we impose is that players possess what is known as *perfect recall*. Loosely speaking, perfect recall means that a player does not forget what she once knew, including her own actions. Figure 7.C.4 depicts two games in which this condition is not met. In Figure 7.C.4(a), as the game progresses, player 2 forgets a move by player 1 that she once knew (namely, whether player 1 chose ℓ or r). In Figure 7.C.4(b), player 1 forgets her own previous move.¹ All the games we consider in this book satisfy the property of perfect recall.

The use of information sets also allows us to capture play that is simultaneous rather than sequential. This is illustrated in Example 7.C.4 for the game of (standard) Matching Pennies introduced in Example 7.B.1.

1. In terms of the formal specification of the extensive form given later in this section, if we denote the information set containing decision node x by $H(x)$, a game is formally characterized as one of perfect recall if the following two conditions hold: (i) If $H(x) = H(x')$, x is neither a predecessor nor a successor of x' ; and (ii) if x and x' are two decision nodes for player i with $H(x) = H(x')$, and if x'' is a predecessor of x (not necessarily an immediate one) that is also in one of player i 's information sets, with a'' being the action at $H(x'')$ on the path to x , then there must be a predecessor node to x' that is an element of $H(x'')$ and the action at this predecessor node that is on the path to x' must also be a'' .

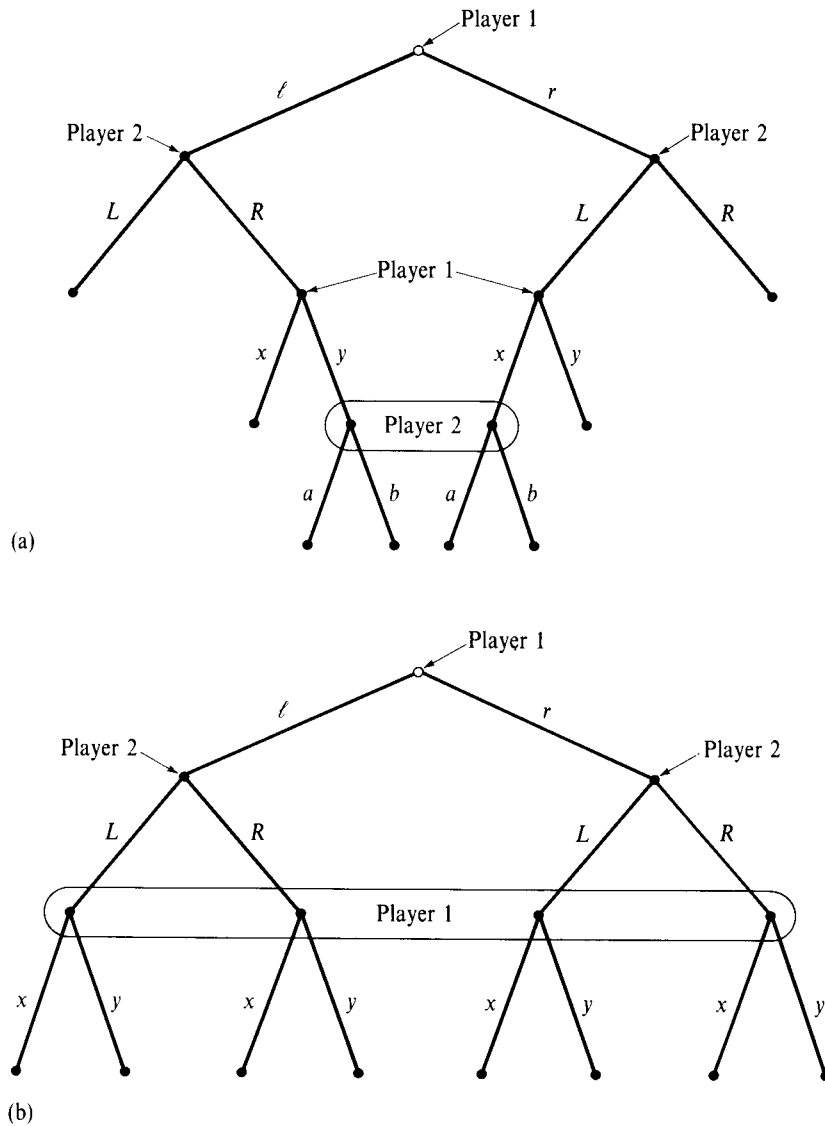


Figure 7.C.4
Two games not
satisfying perfect recall.

Example 7.C.4: The Extensive Form for Matching Pennies. Suppose now that the players put their pennies down simultaneously. For each player, this game is strategically equivalent to the Version C game. In Version C, player 1 was unable to observe player 2's choice because player 1 moved first, and player 2 was unable to observe player 1's choice because player 1 kept it covered; here each player is unable to observe the other's choice because they move simultaneously. As long as they cannot observe each other's choices, the timing of moves is irrelevant. Thus, we can use the game tree in Figure 7.C.3 to describe the game of (standard) Matching Pennies. Note that by this logic we can also describe this game with a game tree that reverses the decision nodes of players 1 and 2 in Figure 7.C.3. ■

We can now return to the notion of a game of perfect information and offer a formal definition.

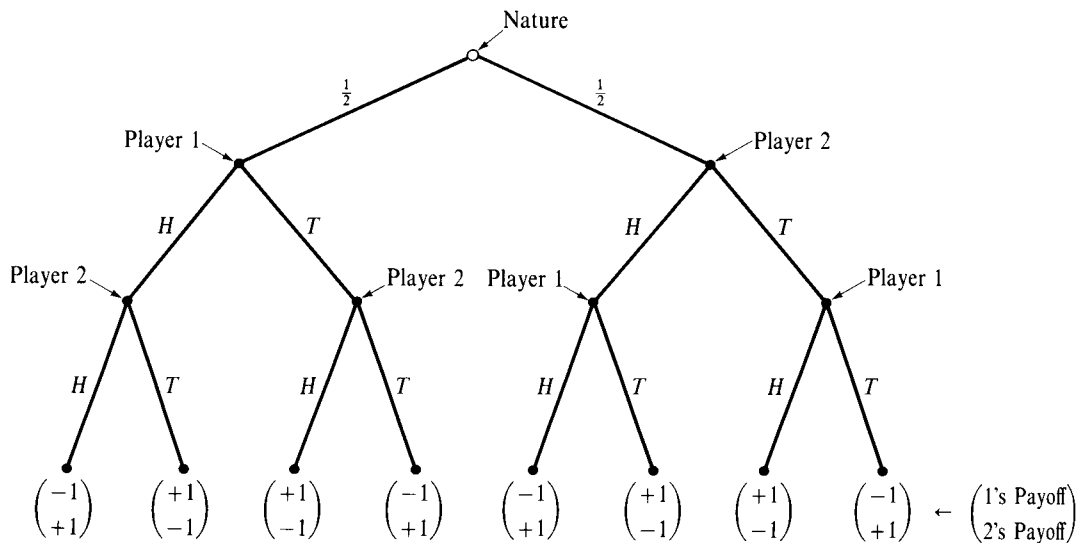


Figure 7.C.5 Extensive form for Matching Pennies Version D.

Definition 7.C.1: A game is one of *perfect information* if each information set contains a single decision node. Otherwise, it is a game of *imperfect information*.

Up to this point, the outcome of a game has been a deterministic function of the players' choices. In many games, however, there is an element of chance. This, too, can be captured in the extensive form representation by including *random moves of nature*. We illustrate this point with still another variation, *Matching Pennies Version D*.

Example 7.C.5: Matching Pennies Version D and Its Extensive Form. Suppose that prior to playing Matching Pennies Version B, the two players flip a coin to see who will move first. Thus, with equal probability either player 1 will put her penny down first, or player 2 will. In Figure 7.C.5, this game is depicted as beginning with a *move of nature* at the initial node that has two branches, each with probability $\frac{1}{2}$. Note that this is drawn as if nature were an additional player who must play its two actions with fixed probabilities. (In the figure, H stands for “heads up” and T stands for “tails up”.) ■

It is a basic postulate of game theory that all players know the structure of the game, know that their rivals know it, know that their rivals know that they know it, and so on. In theoretical parlance, we say that the structure of the game is *common knowledge* [see Aumann (1976) and Milgrom (1981) for discussions of this concept].

In addition to being depicted graphically, the extensive form can be described mathematically. The basic components are fairly easily explained and can help you keep in mind the fundamental building blocks of a game. Formally, a game represented in extensive form consists of the following items:²

2. To be a bit more precise about terminology: A collection of items (i) to (vi) is formally known as an *extensive game form*; adding item (vii), the players' preferences over the outcomes, leads to a *game* represented in extensive form. We will not make anything of this distinction here. See Kuhn (1953) or Section 2 of Kreps and Wilson (1982) for additional discussion of this and other points regarding the extensive form.

- (i) A finite set of nodes $\mathcal{X}$, a finite set of possible actions $\mathcal{A}$, and a finite set of players $\{1, \dots, I\}$.
- (ii) A function $p: \mathcal{X} \rightarrow \{\mathcal{X} \cup \emptyset\}$ specifying a single immediate predecessor of each node x ; $p(x)$ is nonempty for all $x \in \mathcal{X}$ but one, designated as the *initial node* x_0 . The immediate successor nodes of x are then $s(x) = p^{-1}(x)$, and the set of *all* predecessors and *all* successors of node x can be found by iterating $p(x)$ and $s(x)$. To have a tree structure, we require that these sets be disjoint (a predecessor of node x cannot also be a successor to it). The set of *terminal nodes* is $T = \{x \in \mathcal{X}: s(x) = \emptyset\}$. All other nodes $\mathcal{X} \setminus T$ are known as *decision nodes*.
- (iii) A function $\alpha: \mathcal{X} \setminus \{x_0\} \rightarrow \mathcal{A}$ giving the action that leads to any noninitial node x from its immediate predecessor $p(x)$ and satisfying the property that if $x', x'' \in s(x)$ and $x' \neq x''$, then $\alpha(x') \neq \alpha(x'')$. The set of choices available at decision node x is $c(x) = \{a \in \mathcal{A}: a = \alpha(x') \text{ for some } x' \in s(x)\}$.
- (iv) A collection of information sets $\mathcal{H}$, and a function $H: \mathcal{X} \rightarrow \mathcal{H}$ assigning each decision node x to an information set $H(x) \in \mathcal{H}$. Thus, the information sets in $\mathcal{H}$ form a partition of $\mathcal{X}$. We require that all decision nodes assigned to a single information set have the same choices available; formally, $c(x) = c(x')$ if $H(x) = H(x')$. We can therefore write the choices available at information set H as $C(H) = \{a \in \mathcal{A}: a \in c(x) \text{ for } x \in H\}$.
- (v) A function $\iota: \mathcal{H} \rightarrow \{0, 1, \dots, I\}$ assigning each information set in $\mathcal{H}$ to the player (or to nature: formally, player 0) who moves at the decision nodes in that set. We can denote the collection of player i 's information sets by $\mathcal{H}_i = \{H \in \mathcal{H}: i = \iota(H)\}$.
- (vi) A function $\rho: \mathcal{H}_0 \times \mathcal{A} \rightarrow [0, 1]$ assigning probabilities to actions at information sets where nature moves and satisfying $\rho(H, a) = 0$ if $a \notin C(H)$ and $\sum_{a \in C(H)} \rho(H, a) = 1$ for all $H \in \mathcal{H}_0$.
- (vii) A collection of payoff functions $u = \{u_1(\cdot), \dots, u_I(\cdot)\}$ assigning utilities to the players for each terminal node that can be reached, $u_i: T \rightarrow \mathbb{R}$. As we noted in Section 7.B, because we want to allow for a random realization of outcomes we take each $u_i(\cdot)$ to be a Bernoulli utility function.

Thus, formally, a game in extensive form is specified by the collection $\Gamma_E = \{\mathcal{X}, \mathcal{A}, I, p(\cdot), \alpha(\cdot), \mathcal{H}, H(\cdot), \iota(\cdot), \rho(\cdot), u\}$.

We should note that there are three implicit types of finiteness hidden in the formulation just presented. Because we will often encounter games not sharing these features in the economic applications discussed in later chapters, we briefly identify them here, although without any formal treatment. The formal definition of an extensive form representation of a game can be extended to these infinite cases without much difficulty, although there can be important differences in the predicted outcomes of finite and infinite economic models, as we shall see later (e.g., in Chapters 12 and 20).

First, we have assumed that players have a finite number of actions available at each decision node. This would rule out a game in which, say, a player can choose any number from some interval $[a, b] \subset \mathbb{R}$. In fact, allowing for an infinite set of actions requires that we allow for an infinite set of nodes as well. But with this change, items (i) to (vii) remain the basic elements of an extensive form representation (e.g., decision nodes and terminal nodes are still associated with a unique path through the tree).

Second, we have described the extensive form of a game that must end after a finite number of moves (because the set of decision nodes is finite). Indeed, all the examples we have considered so far fall into this category. There are, however, other types of games. For example, suppose that two players with infinite life spans (perhaps two firms) play Matching Pennies repeatedly every January 1. The players discount the money gained or lost at future dates with interest rate r and seek to maximize their discounted net gains. In this game, there are no terminal nodes. Even so, we can still associate discounted payoffs for the two players with every (infinite) sequence of moves the players make. Of course, actually drawing a complete game tree would be impossible, but the basic elements of the extensive form can nonetheless be captured as before (with payoffs being associated with paths through the tree rather than with terminal nodes).

Third, we may at times also imagine that there are an infinite number of players who take actions in a game. For example, models involving overlapping generations of players (as in various macroeconomic models) have this feature, as do models of entry in which we want to allow for an infinite number of potential firms. In the games of this type that we consider, this issue can be handled in a simple and natural manner.

Note that all three of these extensions require that we relax the assumption that there is a finite set of nodes. Games with a finite number of nodes, such as those we have been considering, are known as *finite games*.

For pedagogical purposes, we restrict our attention in Part II to finite games except where specifically indicated otherwise. The extension of the formal concepts we discuss here to the economic games studied later in the book that do not share these finiteness properties is straightforward.

7.D Strategies and the Normal Form Representation of a Game

A central concept of game theory is the notion of a player's *strategy*. A strategy is a *complete contingent plan*, or *decision rule*, that specifies how the player will act in *every possible distinguishable circumstance* in which she might be called upon to move. Recall that, from a player's perspective, the set of such circumstances is represented by her collection of information sets, with each information set representing a different distinguishable circumstance in which she may need to move (see Section 7.C). Thus, a player's strategy amounts to a specification of how she plans to move at each one of her information sets, should it be reached during play of the game. This is stated formally in Definition 7.D.1.

Definition 7.D.1: Let $\mathcal{H}_i$ denote the collection of player i 's information sets, $\mathcal{A}$ the set of possible actions in the game, and $C(H) \subset \mathcal{A}$ the set of actions possible at information set H . A *strategy* for player i is a function $s_i: \mathcal{H}_i \rightarrow \mathcal{A}$ such that $s_i(H) \in C(H)$ for all $H \in \mathcal{H}_i$.

The fact that a strategy is a complete contingent plan cannot be overemphasized, and it is often a source of confusion to those new to game theory. When a player specifies her strategy, it is as if she had to write down an instruction book prior to play so that a representative could act on her behalf merely by consulting that book.

As a complete contingent plan, a strategy often specifies actions for a player at information sets that may not be reached during the actual play of the game.

For example, in Tick-Tack-Toe, player O's strategy describes what she will do on her first move if player X starts the game by marking the center square. But in the actual play of the game, player X might not begin in the center; she may instead mark the lower-right corner first, making this part of player O's plan no longer relevant.

In fact, there is an even subtler point: A player's strategy may include plans for actions that her *own* strategy makes irrelevant. For example, a complete contingent plan for player X in Tick-Tack-Toe includes a description of what she will do after she plays "center" and player O then plays "lower-right corner," even though her own strategy may call for her first move to be "upper-left corner." This probably seems strange; its importance will become apparent only when we talk about dynamic games in Chapter 9. Nevertheless, remember: *A strategy is a complete contingent plan that says what a player will do at each of her information sets if she is called on to play there.*

It is worthwhile to consider what the players' possible strategies are for some of the simple Matching Pennies games.

Example 7.D.1: *Strategies in Matching Pennies Version B.* In Matching Pennies Version B, a strategy for player 1 simply specifies her move at the game's initial node. She has two possible strategies: She can play heads (H) or tails (T). A strategy for player 2, on the other hand, specifies how she will play (H or T) at each of her two information sets, that is, how she will play if player 1 picks H *and* how she will play if player 1 picks T. Thus, player 2 has *four* possible strategies.

Strategy 1 (s_1): Play H if player 1 plays H; play H if player 1 plays T.

Strategy 2 (s_2): Play H if player 1 plays H; play T if player 1 plays T.

Strategy 3 (s_3): Play T if player 1 plays H; play H if player 1 plays T.

Strategy 4 (s_4): Play T if player 1 plays H; play T if player 1 plays T. ■

Example 7.D.2: *Strategies in Matching Pennies Version C.* In Matching Pennies Version C, player 1's strategies are exactly the same as in Version B; but player 2 now only has two possible strategies, "play H" and "play T", because she now has only one information set. She can no longer condition her action on player 1's previous action. ■

We will often find it convenient to represent a profile of players' strategy choices in an I -player game by a vector $s = (s_1, \dots, s_I)$, where s_i is the strategy chosen by player i . We will also sometimes write the strategy profile s as (s_i, s_{-i}) , where s_{-i} is the $(I - 1)$ vector of strategies for players other than i .

The Normal Form Representation of a Game

Every profile of strategies for the players $s = (s_1, \dots, s_I)$ induces an outcome of the game: a sequence of moves actually taken and a probability distribution over the terminal nodes of the game. Thus, for any profile of strategies $(s_1, \dots, s_I)$, we can deduce the payoffs received by each player. We might think, therefore, of specifying the game directly in terms of strategies and their associated payoffs. This second way to represent a game is known as the *normal* (or *strategic*) *form*. It is, in essence, a condensed version of the extensive form.

		Player 2			
		s_1	s_2	s_3	s_4
Player 1	H	-1, +1	-1, +1	+1, -1	+1, -1
	T	+1, -1	-1, +1	+1, -1	-1, +1

Figure 7.D.1
The normal form of
Matching Pennies
Version B.

Definition 7.D.2: For a game with I players, the *normal form representation* Γ_N specifies for each player i a set of strategies S_i (with $s_i \in S_i$) and a payoff function $u_i(s_1, \dots, s_I)$ giving the von Neumann–Morgenstern utility levels associated with the (possibly random) outcome arising from strategies $(s_1, \dots, s_I)$. Formally, we write $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$.

In fact, when describing a game in its normal form, there is no need to keep track of the specific moves associated with each strategy. Instead, we can simply number the various possible strategies of a player, writing player i 's strategy set as $S_i = \{s_{i1}, s_{i2}, \dots\}$ and then referring to each strategy by its number.

A concrete example of a game in normal form is presented in Example 7.D.3 for Matching Pennies Version B.

Example 7.D.3: *The Normal Form of Matching Pennies Version B.* We have already described the strategy sets of the two players in Example 7.D.1. The payoff functions are

$$u_1(s_1, s_2) = \begin{cases} +1 & \text{if } (s_1, s_2) = (\text{H, strategies 3 or 4}) \text{ or } (\text{T, strategies 1 or 3}), \\ -1 & \text{if } (s_1, s_2) = (\text{H, strategies 1 or 2}) \text{ or } (\text{T, strategies 2 or 4}), \end{cases}$$

and $u_2(s_1, s_2) = -u_1(s_1, s_2)$. A convenient way to summarize this information is in the “game box” depicted in Figure 7.D.1. The different rows correspond to the strategies of player 1, and the columns to those of player 2. Within each cell, the payoffs of the two players are depicted as $(u_1(s_1, s_2), u_2(s_1, s_2))$. ■

Exercise 7.D.2: Depict the normal forms for Matching Pennies Version C and the standard version of Matching Pennies.

The idea behind using the normal form representation to study behavior in a game is that a player's decision problem can be thought of as one of choosing her strategy (her contingent plan of action) given the strategies that she thinks her rivals will be adopting. Because each player is faced with this problem, we can think of the players as simultaneously choosing their strategies from the sets $\{S_i\}$. It is as if the players each simultaneously write down their strategies on slips of paper and hand them to a referee, who then computes the outcome of the game from the players' submitted strategies.

From the previous discussion, it is clear that for any extensive form representation of a game, there is a unique normal form representation (more precisely, it is unique up to any renaming or renumbering of the strategies). The converse is not true, however. Many different extensive forms may be represented by the same normal form. For example, the normal form shown in Figure 7.D.1 represents not only the extensive form in Figure 7.C.1 but also the

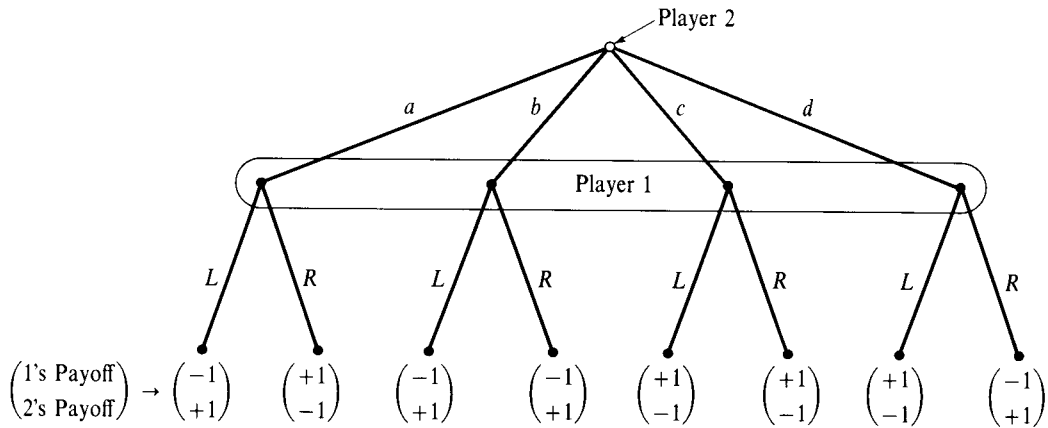


Figure 7.D.2 An extensive form whose normal form is that depicted in Figure 7.D.1.

extensive form in Figure 7.D.2. In the latter game, players move simultaneously, player 1 choosing between two strategies, L and R , and player 2 choosing among four strategies: a , b , c , and d . In terms of their representations in a game box, the only difference between the normal forms for these games lies in the “labels” given to the rows and columns.

Because the condensed representation of the game in the normal form generally omits some of the details present in the extensive form, we may wonder whether this omission is important or whether the normal form summarizes all of the strategically relevant information (as the last paragraph in regular type seems to suggest). The question can be put a little differently: Is the scenario in which players simultaneously write down their strategies and submit them to a referee really equivalent to their playing the game over time as described in the extensive form? This question is currently a subject of some controversy among game theorists. The debate centers on issues arising in dynamic games such as those studied in Chapter 9.

For the simultaneous-move games that we study in Chapter 8, in which all players choose their actions at the same time, the normal form captures *all* the strategically relevant information. In simultaneous-move games, a player’s strategy is a simple non-contingent choice of an action. In this case, players’ simultaneous choice of strategies in the normal form is clearly equivalent to their simultaneous choice of actions in the extensive form (captured there by having players not observing each other’s choices).

7.E Randomized Choices

Up to this point, we have assumed that players make their choices with certainty. However, there is no a priori reason to exclude the possibility that a player could randomize when faced with a choice. Indeed, we will see in Chapters 8 and 9 that in certain circumstances the possibility of randomization can play an important role in the analysis of games.

As stated in Definition 7.D.1, a deterministic strategy for player i , which we now call a *pure strategy*, specifies a deterministic choice $s_i(H)$ at each of her information sets $H \in \mathcal{H}_i$. Suppose that player i ’s (finite) set of pure strategies is S_i . One way for

the player to randomize is to choose randomly one element of this set. This kind of randomization gives rise to what is called a *mixed strategy*.

Definition 7.E.1: Given player i 's (finite) pure strategy set S_i , a *mixed strategy* for player i , $\sigma_i: S_i \rightarrow [0, 1]$, assigns to each pure strategy $s_i \in S_i$ a probability $\sigma_i(s_i) \geq 0$ that it will be played, where $\sum_{s_i \in S_i} \sigma_i(s_i) = 1$.

Suppose that player i has M pure strategies in set $S_i = \{s_{1i}, \dots, s_{Mi}\}$. Player i 's set of possible mixed strategies can therefore be associated with the points of the following simplex (recall our use of a simplex to represent lotteries in Chapter 6):

$$\Delta(S_i) = \{(\sigma_{1i}, \dots, \sigma_{Mi}) \in \mathbb{R}^M: \sigma_{mi} \geq 0 \text{ for all } m = 1, \dots, M \text{ and } \sum_{m=1}^M \sigma_{mi} = 1\}.$$

This simplex is called the *mixed extension* of S_i . Note that a pure strategy can be viewed as a special case of a mixed strategy in which the probability distribution over the elements of S_i is degenerate.

When players randomize over their pure strategies, the induced outcome is itself random, leading to a probability distribution over the terminal nodes of the game. Since each player i 's normal form payoff function $u_i(s)$ is of the von Neumann–Morgenstern type, player i 's payoff given a profile of mixed strategies $\sigma = (\sigma_1, \dots, \sigma_I)$ for the I players is her expected utility $E_\sigma[u_i(s)]$, the expectation being taken with respect to the probabilities induced by σ on pure strategy profiles $s = (s_1, \dots, s_I)$. That is, letting $S = S_1 \times \dots \times S_I$, player i 's von Neumann–Morgenstern utility from mixed strategy profile σ is

$$\sum_{s \in S} [\sigma_1(s_1) \sigma_2(s_2) \cdots \sigma_I(s_I)] u_i(s),$$

which, with a slight abuse of notation, we denote by $u_i(\sigma)$. Note that because we assume that each player randomizes on her own, we take the realizations of players' randomizations to be independent of one another.³

The basic definition of the normal form representation need not be changed to accommodate the possibility that players might choose to play mixed strategies. We can simply consider the normal form game $\Gamma_N = [I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$ in which players' strategy sets are extended to include both pure and mixed strategies.

Note that we can equivalently think of a player forming her mixed strategy as follows: Player i has access to a private signal θ_i that is uniformly distributed on the interval $[0, 1]$ and is independent of other players' signals, and she forms her mixed strategy by making her plan of action contingent on the realization of the signal. That is, she specifies a pure strategy $s_i(\theta_i) \in S_i$ for each realization of θ_i . We shall return to this alternative interpretation of mixed strategies in Chapter 8.

If we use the extensive form description of a game, there is another way that player i could randomize. Rather than randomizing over the potentially very

3. In Chapter 8, however, we discuss the possibility that players' randomizations could be correlated.

large set of pure strategies in S_i , she could randomize separately over the possible actions at each of her information sets $H \in \mathcal{H}_i$. This way of randomizing is called a *behavior strategy*.

Definition 7.E.2: Given an extensive form game Γ_E , a *behavior strategy* for player i specifies, for every information set $H \in \mathcal{H}_i$ and action $a \in C(H)$, a probability $\lambda_i(a, H) \geq 0$, with $\sum_{a \in C(H)} \lambda_i(a, H) = 1$ for all $H \in \mathcal{H}_i$.

As might seem intuitive, for games of perfect recall (and we deal only with these), the two types of randomization are equivalent. For any behavior strategy of player i , there is a mixed strategy for that player that yields exactly the same distribution over outcomes for any strategies, mixed or behavior, that might be played by i 's rivals, and vice versa [this result is due to Kuhn (1953); see Exercise 7.E.1]. Which form of randomized strategy we consider is therefore a matter of analytical convenience; we typically use behavior strategies when analyzing the extensive form representation of a game and mixed strategies when analyzing the normal form.

Because the way we introduce randomization is solely a matter of analytical convenience, we shall be a bit loose in our terminology and refer to all randomized strategies as *mixed strategies*.

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EXERCISES

7.C.1^A Suppose that in the Meeting in New York game (Example 7.B.3), there are two possible places where the two players can meet: Grand Central Station and the Empire State Building. Draw an extensive form representation (game tree) for this game.

7.D.1^B In a game where player i has N information sets indexed $n = 1, \dots, N$ and M_n possible actions at information set n , how many strategies does player i have?

7.D.2^A In text.

7.E.1^B Consider the two-player game whose extensive form representation (excluding payoffs) is depicted in Figure 7.Ex.1.

(a) What are player 1's possible strategies? Player 2's?

(b) Show that for any behavior strategy that player 1 might play, there is a realization equivalent mixed strategy; that is, a mixed strategy that generates the same probability distribution over the terminal nodes for *any* mixed strategy choice by player 2.

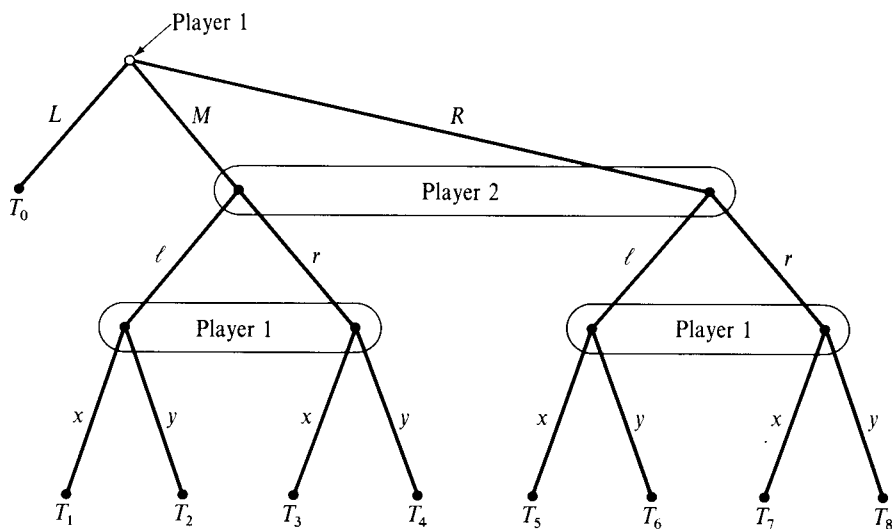


Figure 7.Ex.1

(c) Show that the converse is also true: For any mixed strategy that player 1 might play, there is a realization equivalent behavior strategy.

(d) Suppose that we change the game by merging the information sets at player 1's second round of moves (so that all four nodes are now in a single information set). Argue that the game is no longer one of perfect recall. Which of the two results in (b) and (c) still holds?

Simultaneous-Move Games

8.A Introduction

We now turn to the central question of game theory: What should we expect to observe in a game played by rational players who are fully knowledgeable about the structure of the game and each others' rationality? In this chapter, we study *simultaneous-move* games, in which all players move only once and at the same time. Our motivation for beginning with these games is primarily pedagogic; they allow us to concentrate on the study of strategic interaction in the simplest possible setting and to defer until Chapter 9 some difficult issues that arise in more general, dynamic games.

In Section 8.B, we introduce the concepts of *dominant* and *dominated* strategies. These notions and their extension in the concept of *iterated dominance* provide a first and compelling restriction on the strategies rational players should choose to play.

In Section 8.C, we extend these ideas by defining the notion of a *rationalizable strategy*. We argue that the implication of players' common knowledge of each others' rationality and of the structure of the game is precisely that they will play rationalizable strategies.

Unfortunately, in many games, the set of rationalizable strategies does not yield a very precise prediction of the play that will occur. In the remaining sections of the chapter, we therefore study solution concepts that yield more precise predictions by adding "equilibrium" requirements regarding players' behavior.

Section 8.D begins our study of equilibrium-based solution concepts by introducing the important and widely applied concept of *Nash equilibrium*. This concept adds to the assumption of common knowledge of players' rationality a requirement of *mutually correct expectations*. By doing so, it often greatly narrows the set of predicted outcomes of a game. We discuss in some detail the reasonableness of this requirement, as well as the conditions under which we can be assured that a Nash equilibrium exists.

In Sections 8.E and 8.F, we examine two extensions of the Nash equilibrium concept. In Section 8.E, we broaden the notion of a Nash equilibrium to cover situations with *incomplete information*, where each player's payoffs may, to some extent, be known only by the player. This yields the concept of *Bayesian Nash*

equilibrium. In Section 8.F, we explore the implications of players entertaining the possibility that, with some small but positive probability, their opponents might make a mistake in choosing their strategies. We define the notion of a (*normal form*) *trembling-hand perfect Nash equilibrium*, an extension of the Nash equilibrium concept that requires that equilibria be robust to the possibility of small mistakes.

Throughout the chapter, we study simultaneous-move games using their normal form representations (see Section 7.D). Thus, we use $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$ when we consider only pure (nonrandom) strategy choices and $\Gamma_N = [I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$ when we allow for the possibility of randomized choices by the players (see Section 7.E for a discussion of randomized choices). We often denote a profile of pure strategies for player i 's opponents by $s_{-i} = (s_1, \dots, s_{i-1}, s_{i+1}, \dots, s_I)$, with a similar meaning applying to the profile of mixed strategies σ_{-i} . We then write $s = (s_i, s_{-i})$ and $\sigma = (\sigma_i, \sigma_{-i})$. We also let $S = S_1 \times \dots \times S_I$ and $S_{-i} = S_1 \times \dots \times S_{i-1} \times S_{i+1} \times \dots \times S_I$.

8.B Dominant and Dominated Strategies

We begin our study of simultaneous-move games by considering the predictions that can be made based on a relatively simple means of comparing a player's possible strategies: that of *dominance*.

To keep matters as simple as possible, we initially ignore the possibility that players might randomize in their strategy choices. Hence, our focus is on games $\Gamma_N = [I, \{S_i\}, \{u_i(\cdot)\}]$ whose strategy sets allow for only pure strategies.

Consider the game depicted in Figure 8.B.1, the famous *Prisoner's Dilemma*. The story behind this game is as follows: Two individuals are arrested for allegedly engaging in a serious crime and are held in separate cells. The district attorney (the DA) tries to extract a confession from each prisoner. Each is privately told that if he is the only one to confess, then he will be rewarded with a light sentence of 1 year while the recalcitrant prisoner will go to jail for 10 years. However, if he is the only one not to confess, then it is he who will serve the 10-year sentence. If both confess, they will both be shown some mercy: they will each get 5 years. Finally, if neither confesses, it will still be possible to convict both of a lesser crime that carries a sentence of 2 years. Each player wishes to minimize the time he spends in jail (or maximize the negative of this, the payoffs that are depicted in Figure 8.B.1).

What will the outcome of this game be? There is only one plausible answer: (confess, confess). To see why, note that playing "confess" is each player's best strategy *regardless of what the other player does*. This type of strategy is known as a *strictly dominant strategy*.

		Prisoner 2	
		Don't Confess	Confess
Prisoner 1	Don't Confess	-2, -2	-10, -1
	Confess	-1, -10	-5, -5

Figure 8.B.1
The Prisoner's Dilemma.